

ITU UMC Data Hackathon 2026

Team: High-Five

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1. Overview

1.1 Theoretical Understanding

This project examines Seoul's digital connectivity through three analytical stages and derives policy recommendations on the basis of the results. Digital connectivity is grounded in the Universal Meaningful Connectivity (UMC) framework proposed by the ITU. The project's analyses are interpreted not as a juxtaposition of separate outputs, but within a single metatheoretical framework. Metatheory, as a comprehensive view of the nature of reality, provides philosophical support for research and practice (Allana & Clark, 2018), and offers a basis for specifying the ontological position of each methodology. By explicitly showing the process through which observed phenomena are generalized into concepts at different levels, the project seeks to establish its own validity.

This approach has the advantage of providing not only validity for the project itself, but also a clear basis for application in other contexts. It can offer alternative points of reference for current policy implementation. As evidence-based policy has become widespread, there have been increasing cases in which analytical results are borrowed and applied without sufficient consideration of the institutional and cultural conditions of the original context (Dolowitz & Marsh, 2000). At the same time, critiques of the evidence-based paradigm itself have accumulated. A representative critique is that the narrow evidence-based framework fails to capture the complex, context-dependent, and value-laden character of policymaking (Greenhalgh & Russell, 2009). Accordingly, a transition from evidence-based policy to evidence-informed policy has been discussed (Head, 2010), and the role of evidence is being redefined as guiding policy rather than constituting its systematic foundation.

Rather than diminishing the importance of empirical evidence, this project aims to clarify in detail the conditions and core structures required for applying evidence to other contexts on the basis of a theoretical framework. It explicitly presents the role of each analysis, their interconnections, and the interpretive structure that leads to policy recommendations, while also specifying the key contextual information needed for interpretation. This reflects the case-study character of the project itself, which is distinct from the individual analytical methods.

The individual analytical methodologies are divided into three types. First, the theoretical framework of UMC is applied to Seoul's 25 autonomous districts, which are the actors responsible for policy implementation, in order to measure disparities. Next, a multilevel model is used to separate the contributions of individual factors and district-level factors to individuals' digital connectivity problems. Finally, empirical evidence on Seoul's digital connectivity problems generalized in the preceding analyses is restored to individual phenomena observed through unstructured text. This includes a procedure for abstracting and reconstructing individual experiences in accordance with the conceptual framework of UMC. The concept of UMC proposed by the ITU is compared with aggregated results and actual empirical evidence. Finally, the causal mechanisms through which actual phenomena occur are inferred from the concept. These complex transitions across levels are grounded in the stratified ontology proposed by critical realism, namely the distinction among the empirical, real, and actual domains (Bhaskar, 1975).

The need for this process can be illustrated through a hypothetical situation. Suppose, for example, that observed empirical evidence shows that residential areas in station areas served by express subway lines have higher housing prices than residential areas around stations that are not served by such lines. Apart from the validity of the result, the finding itself is difficult to regard as meaningful knowledge. It may apply only to Seoul. Moreover, making every station served by an express line on the basis of this result is neither politically nor logically feasible. Universally valid knowledge about the world is established through valid inference that goes beyond the generalization of empirical evidence (Bhaskar, 1975). In the preceding situation, one example would be to identify a mechanism whereby the high value of time in Seoul causes express lines that reduce travel time to be reflected in housing prices. The next section explains the advantages that the individual analytical methodologies themselves offer for application in other contexts.

1.2 Understanding the Analytical Methods

The purpose of this section is to explain the three analytical methods and, separately from the metatheoretical framework presented above, to clarify the basis on which each methodology itself can be applied to other contexts. First, empirical patterns are identified through regression-based analysis of the data. The discussion separates the validity of the metalevel structure, which measures and abstracts individual observations and then restores them to individual phenomena while testing whether the UMC concept is also manifested empirically, from the applicability of the individual methodologies.

First, district-level UMC in Seoul is measured to show where disparities exist and along which dimensions. The measurement indicators are based on the conceptual definitions of the six dimensions proposed by ITU (2023), but are operationalized to fit Seoul's data environment and urban characteristics. The actors that actually implement policy are distributed not only at the national level but also across various units, including local governments. Accordingly, this project sets Seoul's 25 autonomous districts as the unit of measurement and analysis so that the analytical unit corresponds to the unit of policy implementation. Because the conceptual definitions and the operational definitions appropriate to the study area are applied separately, indicators can be substituted to fit each city's data environment while maintaining the same six-dimensional structure. The selection of measurement indicators and the results are detailed in Section 3.1.

Next, using a two-level hierarchical linear model (HLM) that includes both individual characteristics and autonomous-district-level UMC infrastructure characteristics, the analysis examines for whom and under what conditions differences in digital usage are most pronounced. The aim is to empirically generalize the causal mechanisms of the digital usage divide. Even within the same administrative area, digital usage can vary substantially according to individuals' educational attainment, age, and household composition. HLM is a standard method for accounting for the hierarchical data structure in which individuals are nested within districts, and the same analytical framework can be applied to other cities by replacing the input variables according to the demographic characteristics and infrastructure conditions of the relevant context. The selection of variables, model specification, and detailed design of cross-level interactions are presented in Section 3.2.

Finally, unstructured text is used to restore individual phenomena that the preceding two analyses cannot capture, to abstract them into higher-level structures appropriate to the UMC concept, and to test the coherence between inferred causal mechanisms and the observed empirical world. Individual cases are not immediately equated with the district-level UMC concept; rather, they pass through a process of abstraction and are compared with the preceding observed results in order to reveal deeper causal mechanisms. The need for this process can be illustrated through a hypothetical situation.

Text analysis has two requirements. First, it must be possible to determine whether individual text cases are related to digital connectivity and to classify them according to the predefined theoretical framework of the six UMC dimensions. Second, the analytical method must not be strongly tied to a particular language. Because the UMC framework is comprehensive, text analysis likewise requires language independence. To satisfy these two conditions, a large language model (LLM) is used as a theory-driven text classification tool. Because an LLM processes meaning at the contextual level rather than at the morpheme level, it is robust to nonstandard expressions in Korean everyday-life text; by directly presenting the UMC dimensions in the prompt, the researcher can control the classification criteria. Methodological transferability is secured insofar as the same prompt structure can be applied to texts in other languages.

Specifically, the measured UMC values from Section 3.1 are combined as the prior distribution with the text classification results as the likelihood to estimate district-level UMC through Bayesian estimation. Then, within the range permitted by the HLM results in Section 3.2, the mechanisms between structure and experience are visualized. Finally, the causal mechanisms between individual posts and the regional UMC concept are inferred. At this stage, three independent layers of inference are designed—abductive, prospective, and path-based—so that each inference agent generates hypotheses without referring to the outputs of the others, after which a higher-level adjudication agent synthesizes them. Separating the inference layers suppresses bias from any single inferential path and makes it possible to observe

convergence and divergence among hypotheses. The prompt design, strategies for reducing misclassification, Bayesian updating, and detailed inference procedures are presented in Section 3.3.

In summary, within the structure of this project, Chapter 2 provides the interpretive context for reading the results in Chapter 3, and Section 3.3 functions as a mediating stage that reconnects the results of Sections 3.1 and 3.2 with experiences in the lifeworld. The three analyses answer different questions, while forming a sequential structure in which the output of one analysis becomes the input for the next analysis (see figure 1). By separating conceptual structure from operational definitions and specifying which elements are replaceable and which are fixed, each methodology establishes the conditions under which the same analytical framework can be applied beyond Seoul (see figure 1).

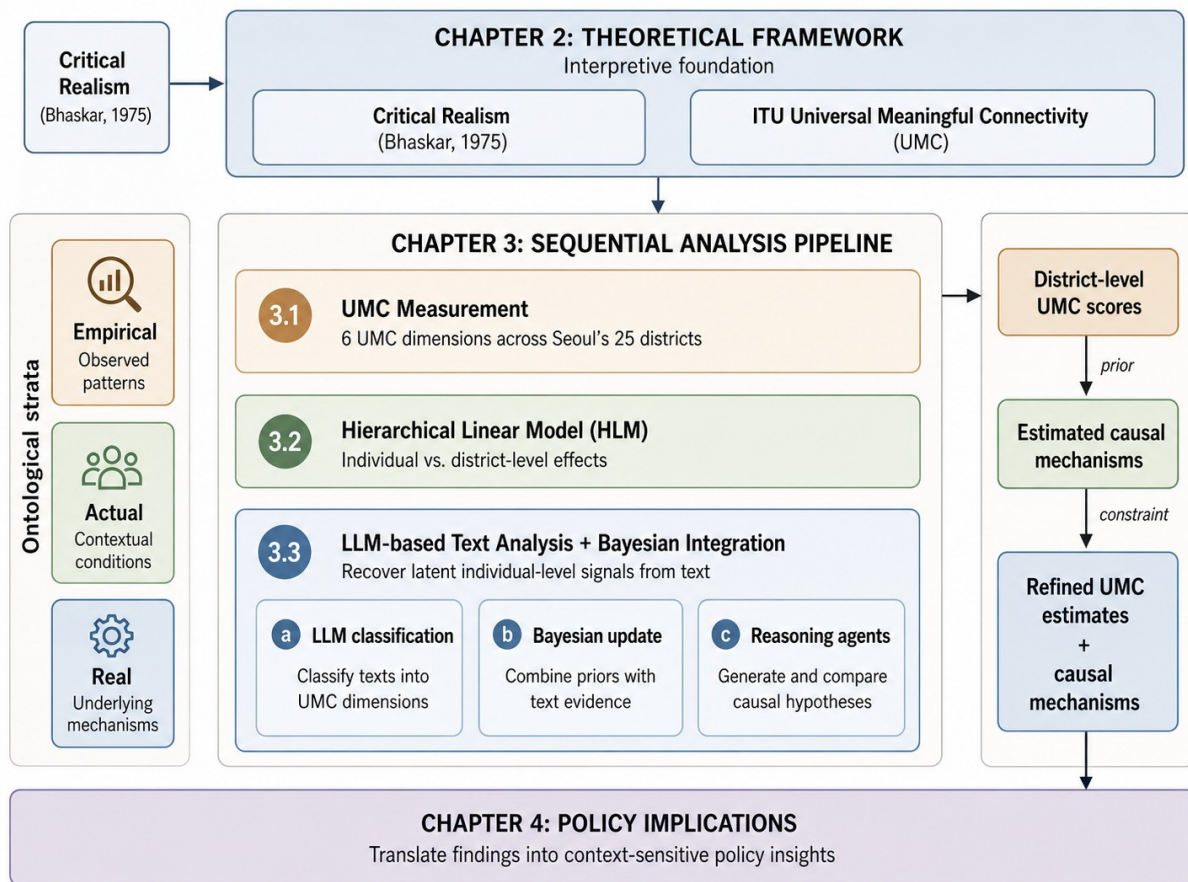


Figure 1. Project Overview

2. Theoretical Foundations and Interpretive Context

This chapter presents the theoretical foundations for the analysis and the interpretive context for Chapter 3. It first reviews the individual and regional factors that affect the digital divide. It then examines the distinction in public policy between a people-based approach and a place-based approach, as well as the background of recent discussions on a place-sensitive approach. Next, it reviews critical realism, the metatheoretical framework of this project, and cases of its application in empirical research. Finally, it examines Seoul's demographic, administrative, and infrastructure conditions in order to provide the contextual information needed to interpret the analytical results.

2.1 Policy

In the design of public policy, the distinction between interventions that target people's characteristics, or a people-based approach, and those that target place conditions, or a place-based approach, is a long-standing debate. People-based policies seek to improve outcomes by directly supporting human characteristics such as individual competency, income, and education. Place-based policies seek to indirectly improve outcomes for residents of a given area by improving that area's infrastructure, institutions, and environmental conditions (Neumark & Simpson, 2015). The two approaches are not mutually exclusive, but because the allocation of policy resources and the pathways of intervention differ, they need to be designed separately from the perspective of policy implementation.

2.1.1 Factors Affecting the Digital Divide: Individual Characteristics and Regional Conditions

In the literature on the digital divide, studies that focus on individual characteristics show that, even within the same region, individual characteristics are a principal factor explaining disparities in digital usage. Hargittai (2010) argued that the divide after internet access—that is, differences in patterns of use—determines substantive inequality. This finding suggests that disparities in digital usage may persist according to socioeconomic status. Chipeva et al. (2018) showed that usefulness is the strongest motivation for technology use across various countries, while also demonstrating that individuals' internal characteristics significantly affect the use of digital devices. Borges et al. (2020), through the case of Angola, reported that habits and proficiency in ICT skills are key factors in the divide prior to the diffusion of devices. Hilbert's (2011) analysis of 18 Latin American countries showed that a gender digital divide exists and that internet use is more active in countries with higher levels of gender equality.

At the same time, evidence has also accumulated that regional environments have additional explanatory power for digital usage even after controlling for individual characteristics. Applying HLM to the United States, Mossberger et al. (2006) showed that disparities in internet access stem from high-poverty neighborhood environments rather than from race itself. This suggests that the structural conditions of a region can independently affect individual outcomes. Through a comparative qualitative case study of Newcastle in the United Kingdom, Crang et al. (2006) showed that the local environment structures the way digital networks are used. Hong et al.'s (2017) analysis of Chinese adults aged 45 and older showed that internet users are concentrated among high-income, highly educated, and urban-resident groups, suggesting that district-level infrastructure and opportunity structures can condition the possibility of use.

2.1.2 Policy Distinctions: People-Based, Place-Based, and Place-Sensitive Approaches

The two types of factors reviewed in the preceding section correspond to the distinction in public policy design between a people-based approach and a place-based approach. People-based policies seek to improve outcomes by directly supporting human characteristics such as individual competency, income, and education. Place-based policies seek to indirectly improve outcomes for residents of a given area by improving that area's infrastructure, institutions, and environmental conditions (Neumark & Simpson, 2015). The two approaches are not mutually exclusive, but because the allocation of policy resources and the pathways of intervention differ, they need to be designed separately from the perspective of policy implementation.

However, the relationship between the two approaches is not a simple parallel. Hindman's (2000) time-series analysis of the United States showed that income, age, and education are stronger predictors than geography, while also reporting a pattern in which disparities become entrenched over time. This implies that individual characteristics and regional structures can interact over time. van Dijk (2020) proposed a sequential model moving from lack of motivation to material access, skills, and substantive use, suggesting that individual factors and structural factors operate cumulatively.

In this context, recent policy and academic literature has been moving away from an either-or choice between people-based and place-based approaches and toward a place-sensitive approach that understands people's experiences within the context of place. One example is the attempt to recognize lived experience and place-based insight as forms of knowledge equivalent to quantitative evidence (OECD, 2025). This approach requires that people-based factors and place-based factors be analytically separated, while also considering how the two combine in specific places.

2.1.3 Metatheoretical Framework: Critical Realism's Stratified Ontology

This project adopts the separation of people-based factors and place-based factors as a basic principle of its analytical design, and grounds the metatheoretical foundation of that design in the stratified ontology of critical realism. According to Bhaskar(1975), the world consists of three domains. The real domain includes structures, mechanisms, and causal powers that exist independently of whether they are observed. The actual domain is the domain in which these structures and mechanisms operate under particular conditions and generate events. The empirical domain refers to the subset of events that have occurred and are observable. Not all events in the actual domain are experienced, nor do observed phenomena represent the entirety of the real domain.

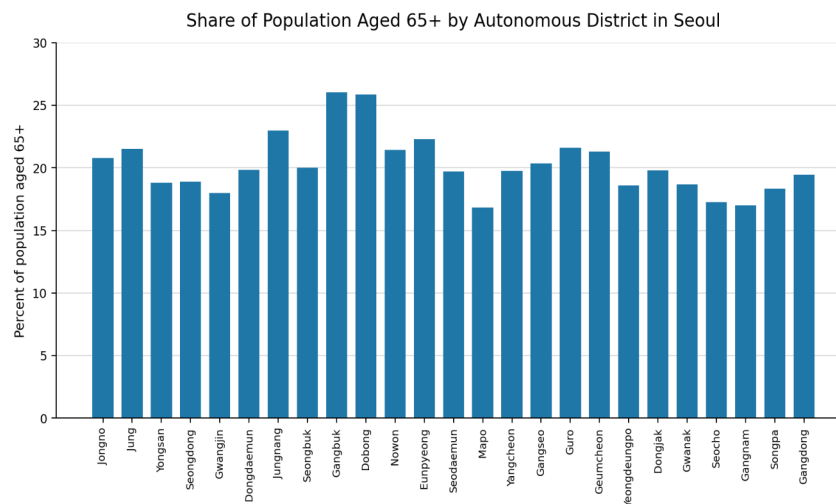
This ontological distinction entails the logic of retrodution(Bhaskar, 1975). Retrodution is a method that infers backward from observed empirical events to the structures and mechanisms that made those events possible. It is distinct from induction and deduction, and asks, "What structure must exist for this phenomenon to be possible?"

2.2 Seoul

2.2.1 Seoul as a City

Seoul, the capital of the Republic of Korea, is a highly dense metropolis in which 20% of the national population resides on 0.6% of the country's total land area. Because of its high density and concentration of technological capacity, Seoul has stood at the forefront of South Korea's digital transformation. Yet a closer look inside Seoul reveals meaningful disparities associated with the demographic structure and economic self-sufficiency of each of Seoul's 25 autonomous districts. This section examines the characteristics of each autonomous district in Seoul and discusses the features of Seoul's population geography.

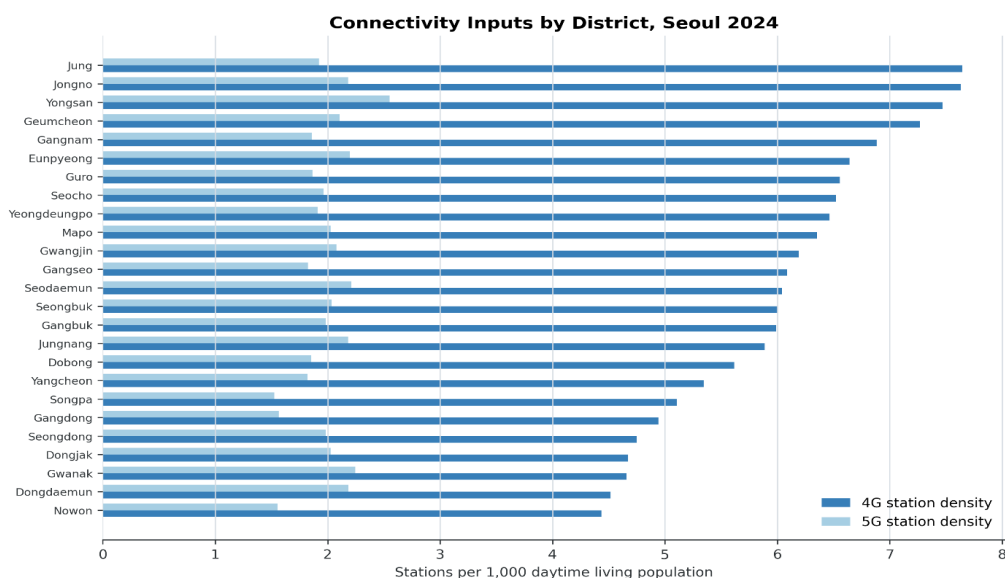
In recent years, Seoul's population geography has been reorganized along two axes: rapid aging and the expansion of single-person households. Northern peripheral districts such as Gangbuk-gu(26.03%) and Dobong-gu(25.85%) already have shares of older adults exceeding 25% and exhibit characteristics of a super-aged society, whereas the Gangnam area and neighborhoods around major universities maintain relatively younger population structures, making inter-district imbalances in age distribution pronounced. These demographic differences, combined with each district's average income level and economic vitality, form the structural backdrop that produces the differences in Affordability and Devices capabilities identified in Section 3.1 of this study.



In addition, Seoul's policy implementation system operates through organic cooperation between the metropolitan government, the Seoul Metropolitan Government, and the 25 lower-level autonomous districts. The broad direction of policy follows a top-down approach established by Seoul City Hall, which plays a central role in building metropolitan-scale infrastructure and setting standard guidelines for digital inclusion. At the same time, Seoul also operates bottom-up decision-making structures, including the citizen participatory budgeting system and residents' associations, through which digital inconveniences experienced by local residents in everyday life are directly reflected in policy. Based on these metropolitan plans, autonomous districts finalize budgets through local council deliberation and ordinance enactment, and ultimately deliver close-to-resident services to citizens through neighborhood community service centers. In this sense, Seoul's policies for reducing the digital divide can be understood as being realized through a multilayered process that combines the administrative capacity of each autonomous district with citizen participation, rather than through uniform central execution.

2.2.2 Current Status of Connectivity Gaps and Related Laws and Policies

The current state of digital connectivity in Seoul clearly reveals the limits of privately led infrastructure deployment through disparities in radio station infrastructure density across autonomous districts. According to Korea Radio Promotion Association statistics on radio stations for 2020-2025, the core infrastructure determining Seoul's communication quality, namely base stations and mobile repeaters, is overwhelmingly concentrated in particular areas where profitability is high and traffic is concentrated. Southeastern districts such as Gangnam-gu(base stations 19,190; mobile repeaters 2,711) and Songpa-gu(base stations 14,751) possess dense networks, whereas northern peripheral districts such as Dobong-gu(base stations 4,008) and Gangbuk-gu(base stations 4,700) show a gap of up to 4.8 times in the number of base stations. In particular, for simple radio stations used for small businesses and Internet of Things(IoT) connectivity, Gangnam-gu(24,236) has about 20 times as many as Dobong-gu(1,224), suggesting that a rich-get-richer and poor-get-poorer pattern in the digital ecosystem across autonomous districts has already become entrenched at the level of physical infrastructure.



To correct these regional imbalances in private infrastructure and guarantee citizens' basic right to communication, Seoul made the policy decision to step forward directly as a facilities-based telecommunications service provider as an administrative alternative. Under the 2025 amendment to the Telecommunications Business Act, Seoul became the first local government to secure the status of a telecommunications service provider, and it is expanding public Wi-Fi and IoT infrastructure by directly connecting more than 5,000 km of its own fiber-optic cable network to underserved areas where private telecommunications companies have invested insufficiently because of profitability concerns. Seoul currently operates more than 34,000 public Wi-Fi units, the largest scale in the country; these are widely installed on streets, in parks, markets, welfare facilities, and city buses, generating an annual reduction of approximately KRW 200 billion in citizens' communication costs. In particular, mobile Wi-Fi installed on roughly 7,000 city buses has become a form of digital infrastructure that citizens experience most directly in everyday life. Through the expansion of public Wi-Fi, Seoul plans to reduce disparities in information access and ease the household burden of communication costs amounting to hundreds of billions of won each year. It also seeks to establish a sustainable virtuous cycle by lowering dependence on privately leased networks and reinvesting the secured resources in projects that benefit citizens.

Policy responses to address these infrastructure gaps have moved beyond physical network expansion and increasingly focus on strengthening Affordability by lowering citizens' actual cost burden. The central and local governments operate the Smart Choice platform to help users understand their own communication consumption patterns and make rational choices. Smart Choice provides integrated services such as checking handset subsidies, recommending customized rate plans, and searching for unclaimed refunds, thereby reducing information asymmetry and encouraging reductions in household communication costs.

The recent reform of 5G rate plans is especially significant in that it shifted a market structure centered on high-priced plans toward one centered on low- and mid-priced plans. The lowest 5G price tier, which had previously remained in the KRW 40,000-50,000 range, was lowered to the KRW 30,000 range, and intermediate plan tiers between 30 and 100GB were newly and more finely introduced to correspond to actual data usage. This reform addressed the structural contradiction in which users paid high costs for data they did not use, and by 2024 it had led to a concrete policy outcome in which more than 6 million subscribers switched to low- and mid-priced plans.

Customized dedicated rate plans for vulnerable groups and specific age groups have also been substantially strengthened. For young people (age 34 and below), the three major mobile carriers in the Republic of Korea expanded the amount of data provided by up to twice that of ordinary plans at the same price, taking into account high data demand among this group, and broadened the eligible population by raising the upper age limit for subscription from age 29 to 34. In addition, for older adults (age 65 and above), LG U+ launched

dedicated rate plans segmented by age group(65, 70, 80, and so on) that are up to 20% cheaper than ordinary plans, reflecting older adults' relatively lower data usage and higher price sensitivity. Furthermore, in line with the trend toward non-face-to-face subscription, online-only rate plans have expanded choice by introducing no-contract plans that are approximately 30% cheaper than ordinary plans when users subscribe through online channels without visiting a retail store.

As a result, Seoul is building a multifaceted inclusion policy that guarantees universal digital basic rights for all citizens through a three-part framework: information transparency through Smart Choice, the rational rate structures of the three major mobile carriers, and Seoul's direct provision of public infrastructure.

In addition, the most significant change in Seoul's administrative services in 2026 is the unification and upgrading of mobile platforms. Seoul will terminate the previously fragmented Seoul Citizen Card and Seoul Wallet services as of March 31, 2026, and consolidate their functions into the integrated mobile platform Seoul ON. By enabling citizens to address everything from the use of public facilities such as libraries and sports facilities to verification of disability or national merit status within a single app, this change lowers the entry barrier to digital administration created by service fragmentation. In particular, Seoul ON enhances administrative efficiency by strengthening the function that verifies eligibility for vulnerable-group status in real time through linkage with public MyData, without requiring separate document submission.

Furthermore, Seoul is pursuing an integrated inclusion policy centered on the AI Digital Learning Center so that infrastructure expansion and service integration lead to genuine improvements in citizens' practical usage capabilities. Using nationwide digital learning center sites, including 10 sites in Seoul, the city provides ongoing face-to-face education for older adults and digitally vulnerable groups, and offers not only basic internet-use instruction but also life-oriented AI experience education and education to prevent digital dysfunction. In particular, it is closing educational blind spots by expanding "visiting 1:1 education" for people such as those with severe disabilities who have mobility difficulties, and by establishing a level-specific customized education system linked across online and offline formats. This approach reflects the judgment that even when hardware(infrastructure) and software(administrative services) are in place, gaps will not be resolved if people lack the human capabilities needed to use them. In this way, Seoul is substantively realizing Universal Meaningful Connectivity(UMC) through a three-part framework: securing infrastructure stability based on law, innovating the service delivery system through Seoul ON, and improving users' capabilities through education.

3. Analysis

The three analyses in this chapter are sequentially connected. First, the study constructs a UMC index by autonomous district in Seoul to measure regional disparities(3.1). It then uses these results as district-level variables to analyze the effects of individual and district characteristics on individuals' digital connectivity(3.2). Finally, it derives concrete cases not captured by the quantitative analysis through digital platform text. This sequence consists of measurement, causal identification, and phenomenological supplementation, and Section 3.4 synthesizes the results of the three analyses.

3.1 Regional Disparities in UMC in Seoul

3.1.1 Measurement Indicators and Data

The measurement indicators constructed in this project are based on the conceptual definitions of the six dimensions of UMC. These dimensions are classified as Connectivity, Available for Use, Affordability, Devices, Skills, and Safety, and the individual measurement indicators were specified according to two core components that simultaneously consider meaningfulness and universality(ITU, 2023). The measurement indicators for the six dimensions comprising Seoul's UMC used in this analysis are shown in <Table 1>.

In selecting measurement indicators, the analysis comprehensively considered alignment with conceptual definitions, Seoul's digital technology environment, and data availability. The analysis used data from two time points, 2023 and 2024, the most recent years for which data could be secured. National administrative data and related sources from these two time points were used. However, because of limitations in the timing of data provision, the Devices, Skills, and Safety dimensions applied the same values for both time points based on 2023 survey data.

For Connectivity, measurement indicators were calculated using mobile communication radio station data provided by Korea Radio Promotion Association and living population data from Seoul's administrative data. According to ITU(2023), Connectivity can be measured based on the population residing within network coverage. In Korea, however, the separation between workplace and residence is often pronounced, so the area of residence may differ from the area where individuals actually access the digital environment. Accordingly, this study devised an indicator more appropriate to the Korean context by calculating 4G and 5G network coverage using daytime floating population data rather than simple resident population. Specifically, the numbers of 4G base stations and 5G base stations were each converted into the number of base stations per 1,000 daytime living population at the enumeration-district level, and the final indicator used the average of these values by autonomous district.

Available for Use is the dimension that measures the practical usability of digital services. Monthly mobile data usage per household within each area, provided by SKT communication information, and online service usage days in four categories, finance, shopping, video, and delivery, were aggregated monthly by autonomous district as population-weighted averages, and the 12-month average values were used. In addition, data on public Wi-Fi registration status from autonomous district governments were used to calculate and include the number of public Wi-Fi APs per 1,000 total population. Public Wi-Fi density functions as an especially important digital access pathway for older adults whose home internet access environment is limited(Van Dijk, 2020).

Affordability was constructed following the two-axis structure of the ITU ICT Development Index 2023(ITU, 2024). For the purchasing power(capacity) side, the analysis used average monthly household income by autonomous district provided by Seoul's Commercial District Analysis Service. For the price burden(burden) side, it used the share of bill arrears within the most recent three months from SKT communication information. The arrears rate was calculated as a population-weighted average by administrative dong, sex, and age group, and then aggregated to the autonomous-district level; it is a reverse-coded indicator, meaning that higher values indicate lower economic accessibility.

Devices measures the possession and diversity of use of digital devices. The weighted averages by autonomous district were calculated for the number of core digital device types owned within households

and the number of device types actually used by individuals, based on the Seoul Citizens' Digital Competency Survey. According to ITU(2023), possession and use are interpreted differently depending on device characteristics, such as mobile phones and computers. In the Korean domestic context, whether a computer is owned can also be an indicator of digital connectivity. In Seoul, where smartphone penetration is effectively saturated, ownership of auxiliary devices such as desktops, laptops, and tablets acts as a differentiating factor in digital usage capability(Van Deursen & Helsper, 2015).

Digital Skills comprises four subdomains: Basic, Applied, Advanced, and Cyber activity. Basic competency measures the basic ability to operate smart devices; Applied competency measures the ability to use digital services; Advanced competency measures proficiency in smart-device activities; and Cyber activity competency measures the ability to act in cyberspace. The raw score for each subdomain was divided by its maximum possible score and converted to a 0-1 ratio, after which weighted averages were calculated by autonomous district.

Safety comprises two subindicators: digital security behavior and security awareness. Security behavior measures actual security practices, such as password management, software updates, and use of two-factor authentication, while security awareness measures the level of recognition of the importance of personal information protection. Each subindicator was converted to a ratio relative to its maximum possible score, and weighted averages were then calculated by autonomous district.

Table 1. UMC Dimension Indicators

Dimension	Indicators	Data Source	Note
Connectivity	station_density_4g, station_density_5g, download_speed	Spectrum Resource Mgmt System; NIA Communication Quality	Per 1,000 daytime living population; CQ uses 2022-2024 average
Available for Use	mobile_data_usage, online_service_days, wifi_density	SKT Telecom Data; Local Administrative Wi-Fi License Data	Population-weighted district average; Wi-Fi per 1,000 total population
Affordability	avg_income, delinq_rate (reverse)	Seoul Commercial Analysis Service; SKT Telecom Data	ITU IDI capacity + burden; delinq = reverse-coded
Devices	device_home_core, device_use_core	Seoul Citizens' Digital Competency Survey 2023	Home: excluding smartphones; Use: including smartphones
Digital Skills	Basic / Applied / Advanced / Cyber activity	Seoul Citizens' Digital Competency Survey 2023	-
Safety	safety_behavior, safety_awareness	Seoul Citizens' Digital Competency Survey 2023	Security practices (7 items) + awareness (3 items)

3.1.2 Indexing the Measurement Indicators

All raw indicators were normalized to the [0, 1] interval through Min-Max scaling across the 25 autonomous districts. The normalized value x^* is calculated as shown in Equation 1.1.

Positive indicator: $x^* = (x - \min(x)) / (\max(x) - \min(x))$

Reverse indicator: $x^* = (\max(x) - x) / (\max(x) - \min(x))$

Computed within each year across Seoul's 25 autonomous districts; composite scores are relative, not absolute UMC levels.

At this stage, reverse-direction indicators, for which larger values indicate worse outcomes, were transformed so that all indicators had a consistent direction. Next, the normalized indicators within each dimension were combined using an equally weighted arithmetic mean to calculate dimension-level scores. Equal weights were used in the analysis to minimize arbitrary researcher judgment in the absence of an agreed standard for the relative importance of the six dimensions (ITU, 2024). The resulting composite index ranges from 0 (lowest) to 1 (highest), and was calculated independently for 2023 and 2024 for all 25 autonomous districts in Seoul. A key point for interpretation is that this composite index represents each district's relative position among other autonomous districts in Seoul, rather than an absolute level of regional UMC.

3.1.3 Characteristics of the UMC Index by District

The 2024 UMC composite index for the 25 autonomous districts ranged from 0.279 (Jungnang-gu) to 0.695 (Seocho-gu), indicating an approximately 2.5-fold gap between the highest and lowest districts. Despite the common perception that Seoul is a city with high internet penetration, a substantial relative digital connectivity divide was observed at the lower administrative-unit level. The top five autonomous districts were Seocho-gu (0.695), Yeongdeungpo-gu (0.686), Mapo-gu (0.665), Seodaemun-gu (0.615), and Yongsan-gu (0.603). The bottom five were Jungnang-gu (0.279), Dobong-gu (0.303), Gangbuk-gu (0.380), Guro-gu (0.389), and Nowon-gu (0.389). Higher-ranked districts were generally located in the central city or the Gangnam area, benefiting from dense commercial activity, high income levels, and abundant private digital infrastructure, whereas lower-ranked districts were concentrated in the northern periphery and the southwest.

Examining the spatial distribution of the composite index (Figure 1), autonomous districts with high UMC index values were concentrated in two areas. These were the Han River and central-city corridor running from Seocho to Yeongdeungpo and Mapo, and the traditional downtown area centered on Yongsan, Jongno, and Jung-gu. By contrast, the northern periphery (Gangbuk-gu, Dobong-gu, Nowon-gu) and the southwest (Guro-gu, Geumcheon-gu) scored below the city average. This spatial distribution was associated with the distribution of socioeconomic resources in Seoul. In other words, it suggests that the digital divide may be closely linked to structural spatial inequality in the city.

At the same time, the dimension-level analysis found that no autonomous district performed uniformly well across all six dimensions (Figure 3). For example, Eunpyeong-gu in northern Seoul ranked high in the Connectivity dimension, with a score of 0.781, but remained in the lower tier in the Affordability dimension, with a score of 0.328. Similarly, Gwanak-gu ranked first in the Digital Skills dimension, with a score of 0.795, but fell into the lowest tier for Affordability, at 0.152. These multidimensional inter-district inequalities indicate that policy-relevant dimension-specific vulnerabilities cannot be captured by the composite ranking alone.

In conclusion, as shown in Figure 4, this project provisionally classifies the four spatially clustered autonomous districts in the lowest group of the composite index (Jungnang-gu, Dobong-gu, Gangbuk-gu, and Nowon-gu) as Digital Deserts. These districts commonly exhibit clear deficits in the Devices and Safety dimensions, and Affordability also falls below the Seoul average. Meanwhile, in the Connectivity dimension, some districts (Jungnang-gu 0.630, Gangbuk-gu 0.562) exceed the Seoul average, indicating that the core divide may stem from insufficient human and economic capabilities to use physical network infrastructure rather than from the absence of that infrastructure itself. This finding is examined more systematically in

Section 3.2 by separating and testing the relative contributions of individual-level and district-level variables in a multilevel analysis.

In addition, geographically adjacent areas displayed similar patterns in some dimensions of UMC. <Figure 4> shows the spatial autocorrelation of districts by dimension. The fact that spatially clustered autonomous districts show similar patterns suggests the possibility of establishing a metropolitan governance system to address Digital Deserts.

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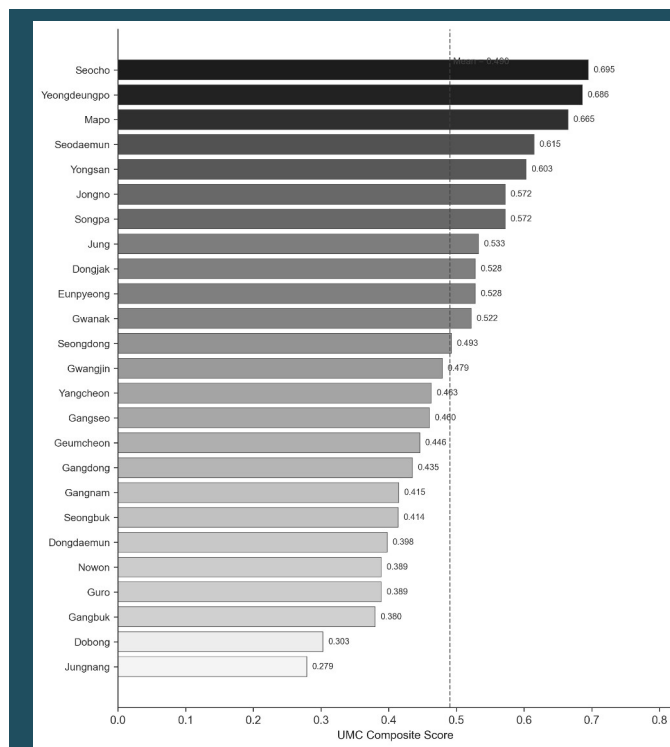


Figure 2. UMC Composite Scores by District

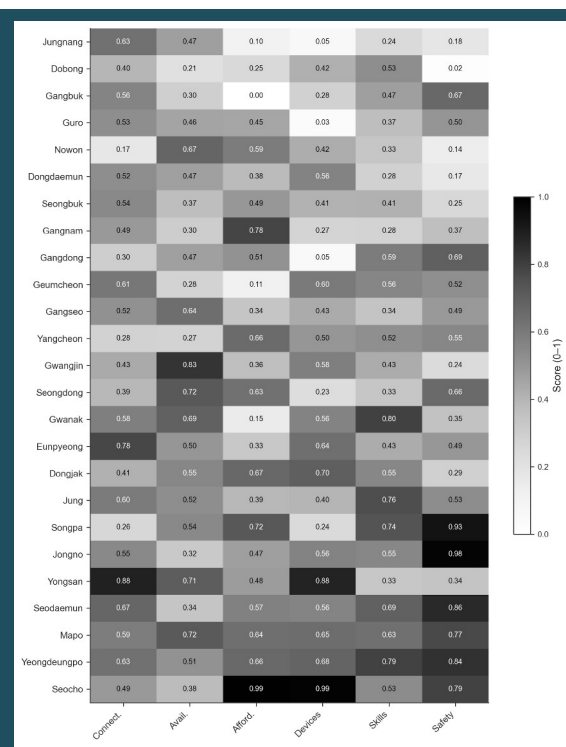


Figure 3. Heatmap of UMC Dimension Scores



Figure 4. UMC Dimension Scores

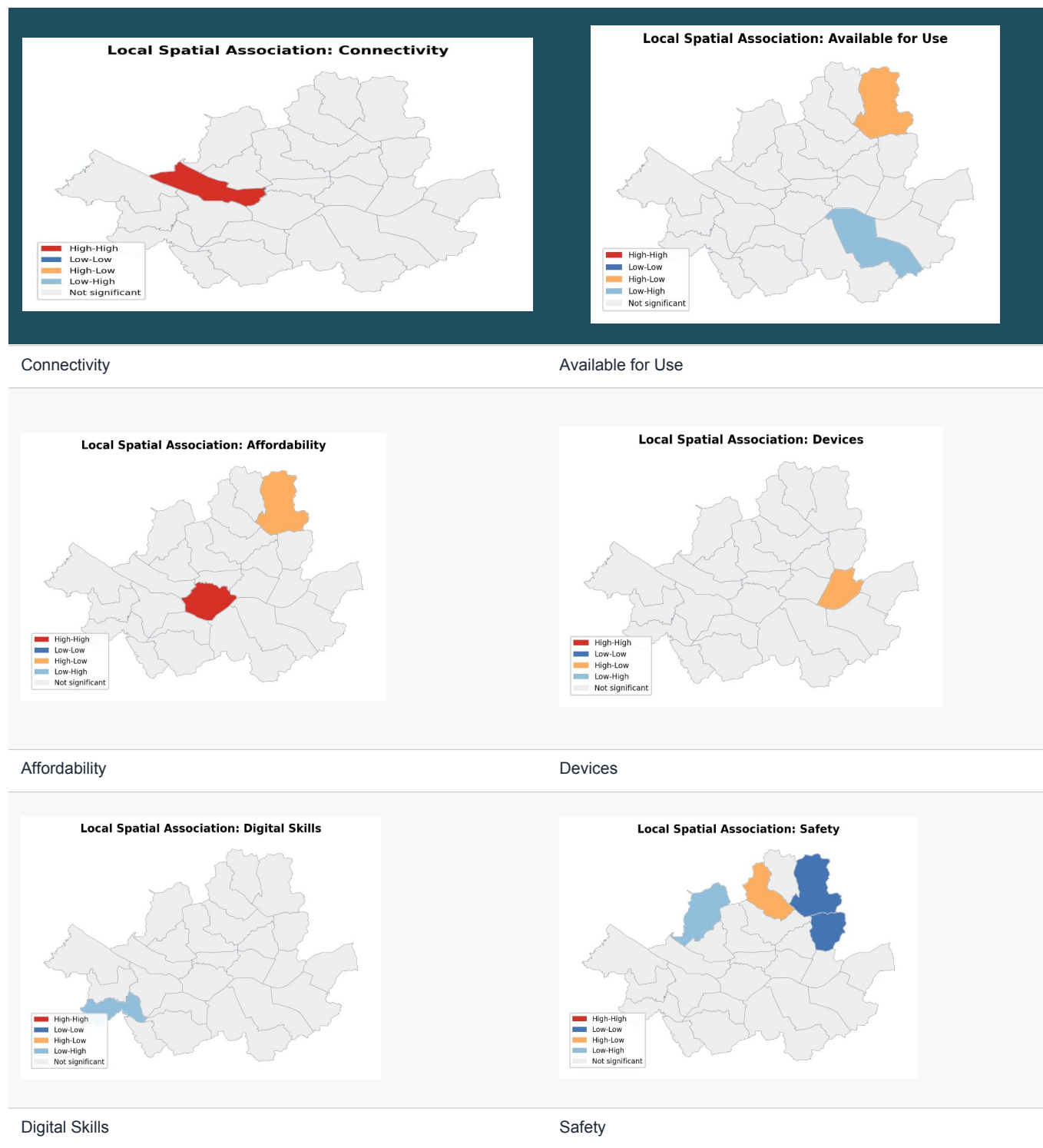


Figure 5. Spatial autocorrelation by UMC level across Seoul's 25 autonomous districts

3.2 Individual and District Characteristics and Digital Connectivity

The data used in this study have a hierarchical structure in which individuals (Level 1) are clustered within autonomous districts (Level 2). Here, Level 2 corresponds to each individual's place of residence. In this structure, conventional OLS regression can violate the assumption of independent observations,

underestimate the standard errors of district-level predictors, and inflate Type I error (Raudenbush & Bryk, 2002). Therefore, this study adopted a two-level hierarchical linear model (HLM) that decomposes variance into within-district and between-district components across Seoul's 25 autonomous districts and allows individual-level and district-level variables to be entered simultaneously.

The Seoul Survey, used in the second analysis, is administrative data that the Seoul Metropolitan Government has collected annually since 2003 from 20,000 households. Data collection is conducted every August-September through a combination of non-face-to-face and in-person interviews.

3.2.1 Variable Construction

First, the dependent variable in the analysis, the digital usage score, is the summed score of eight items on digital service usage frequency among the Seoul Survey response items. The eight items consist of living information services, e-commerce services, financial transaction services, public services, communication services, leisure services, education and learning services, and mobile payment. These are not basic acts of connection such as phone calls or text messaging, but rather substantive additional uses mediated by digital devices, and were therefore judged to be indicators suited to Seoul's advanced digital environment. Each item was transformed from its original four-point ordinal scale into a 0-10 continuous scale, yielding a continuous variable ranging from 0 to 80 points. A higher score means that an individual uses a more diverse range of digital services and can freely use desired services.

The independent variables in this analysis are divided into individual-characteristic variables and district-characteristic variables (Table 2). The individual-characteristic variables consist of sex, age, occupation, educational attainment, income level, and disability status, all of which may affect individual digital usage. Educational attainment was divided into four categories: middle school completion or below, high school completion, college graduation (reference category), and graduate school or above. Occupation was divided into four categories: managers and professionals, clerical workers (reference category), production and technical workers, and other (students, homemakers, and unemployed persons). Housing type was classified into four categories: apartment (reference category), detached house, multiplex and multi-household housing, and other. Age and income were grand-mean centered to improve the interpretability of the intercept. In addition, a single-person household dummy, a disability-status dummy, and a year dummy (reference: 2023) were included.

As Level 2 variables, the analysis used three UMC dimension scores constructed in Section 3.1 that represent exogenous factors determining the local digital environment: Connectivity, Available for Use, and Affordability. This choice reflects that Devices, Digital Skills, and Safety are aggregated to the autonomous-district level from individual responses in the Seoul Citizens' Digital Competency Survey. To avoid the resulting endogeneity problem, these three dimensions were excluded from the main analysis, and dimensions judged more appropriate for district-level universal meaningful connectivity were selected. District average income and the aging rate were included as control variables, and all Level-2 variables were standardized to the [0, 1] range using Min-Max normalization, as in Section 3.1.

The analysis proceeds in four steps. After estimating an unconditional model and calculating the intraclass correlation coefficient (ICC) to confirm the justification for multilevel analysis, the study sequentially estimates M1, which includes individual-level variables; M2, which adds autonomous district-level variables; and M3, which includes cross-level interactions. The cross-level interactions are constructed as products of UMC infrastructure variables (Connectivity, Available for Use, and Affordability) and vulnerable-group indicators, in order to test whether the level of digital infrastructure in an autonomous district operates differently for the digital usage of vulnerable groups.

3.2.2 Analysis Results

The sample used in the analysis consists of a total of 10,000 Seoul Survey respondents from 2023-2024, with an average of 400 sampled from each of the 25 autonomous districts. <Table 3> presents descriptive statistics for the main variables. The mean value of individual digital connectivity was 60 out of 80 points, clearly demonstrating the high level of digital connectivity among Seoul residents. Although the dependent

variable may exhibit negative skewness, the analysis was conducted using the scale-linearized raw scores, given the institutional standard scale and a sample size approaching 10,000.

Table 4. presents the sequential estimation results of the multilevel models. The intraclass correlation coefficient (ICC) of the unconditional model was 0.49%, indicating that the share of between-district variance itself was not large. However, the design effect, calculated in light of the average sample size of 400 persons per autonomous district, was 2.94, exceeding the threshold of 2.0 and confirming the validity of using a multilevel model (Hox et al., 2018). Nevertheless, taken together, the small number of 25 autonomous districts and the very small share of variance explained by autonomous districts suggest that district-level connectivity-related characteristics may have an extremely limited effect on individual digital connectivity. These results may indicate that, because Seoul has excellent digital connectivity and a high level of digital competency, district-level connectivity infrastructure does not exert a measurable effect. In addition, because the Seoul Survey records individuals' residential districts, those districts may not correspond to the areas where individuals are directly affected by connectivity.

Among individual characteristics, the association with educational attainment was the most pronounced. Education of middle school completion or below was associated with a digital usage score approximately 22.6 points lower than that of college graduates. This is a substantial gap, corresponding to 28% of the 80-point scale. High school graduates also scored 7.3 points lower than college graduates, whereas those with graduate school education or above scored 2.9 points higher, confirming a clear positive relationship between educational attainment and digital usage capacity. Age also had a significant effect. For each one-year increase from the mean age, the score for individual digital connectivity decreased by 0.35 points, and among older adults aged 65 or above, an additional 6.0-point decline was observed apart from the age effect. A pronounced decline in digital connectivity appeared among those aged 65 or above. In terms of occupation, managers and professionals scored 2.3 points higher than clerical workers, indicating that higher-income occupations tended to have higher levels of digital usage. In terms of the residential environment, residents of detached houses scored 2.5 points lower than apartment residents, and residents of multiplex or multi-household housing scored 1.8 points lower, which suggests that

Table 4. Variable Descriptions

Variable	Description	Level	Scale	Source
digital_use_score	Digital service usage score (sum of 8 items)	L1	0–80 (continuous)	Seoul Survey 2023–24
age_c	Age (grand-mean centered, M = 49.3)	L1	years	Seoul Survey
female	Female (ref: male)	L1	0/1	Seoul Survey
Edu	Education: ≤ middle school / high school / graduate school or higher (ref: college)	L1	0/1 each	Seoul Survey
Income	Monthly income (grand-mean centered)	L1	10K KRW	Seoul Survey
job	Job: professional / blue-collar / other (ref: white-collar)	L1	0/1 each	Seoul Survey
disabled_bin	Disability status	L1	0/1	Seoul Survey
hh_size_1	Single-person household	L1	0/1	Seoul Survey
housing_detached / _multi / _other	Housing: detached / multi-family / other (ref: apartment)	L1	0/1 each	Seoul Survey
year	Year 2024 dummy (ref: 2023)	L1	0/1	Seoul Survey

elderly	Older adults (≥ 65)	L1	0/1	Seoul Survey
low_edu	Lower educational attainment ($\leq$ middle school)	L1	0/1	Seoul Survey
connectivity	Connectivity dimension score (normalized)	L2	0–1	UMC Part 1
wifi_total_n	Public Wi-Fi density per 1K population (normalized)	L2	0–1	Local Admin License

This finding can be interpreted as reflecting not only income-level differences by housing type but also differences in internet connectivity conditions by building type. Single-person households also scored 1.8 points lower than multi-person households. Among other individual characteristics, women scored 2.6 points higher than men. Women's higher digital connectivity relative to men may reflect their greater responsiveness to various digital services in Korea. (Ref) These results should be interpreted with attention to country-specific sociocultural contexts. Respondents with disabilities scored 3.1 points higher, which appears attributable to the characteristics of the sample.

At the autonomous district level, the UMC infrastructure dimensions were generally not significant as main effects. In Model 2, none of the three dimensions reached statistical significance, and the likelihood-ratio test likewise indicated no significant improvement over Model 1. However, in Model 3, which included cross-level interactions, the Infrastructure dimension showed a significant negative effect. This indicates the presence of differential effects across groups rather than an average infrastructure effect.

The central finding of this analysis emerged in the cross-level interactions. The interaction between public Wi-Fi density and the older adult dummy showed a significant positive effect of 3.03 points. This means that public Wi-Fi infrastructure provides relatively greater benefits to older adults than to non-older adults. Residents with lower educational attainment living in autonomous districts with higher levels of digital infrastructure recorded markedly higher digital usage scores than residents with lower educational attainment in areas with lower infrastructure levels. This finding suggests that infrastructure investment may have a disproportionately large effect in reducing the digital divide among educationally vulnerable groups.

At the individual level, the interactions confirmed a pattern of double vulnerability. Older adults living alone scored an additional 4.0 points lower than older adults not living alone, and the negative effect of lower educational attainment became larger with age. These findings show that compound vulnerability factors intensify the digital divide more than any single vulnerability factor.

Table 5. Descriptive Statistics

Variable	N	Mean	Standard deviation (SD)	Minimum (Min)	Maximum (Max)
Digital usage score	10,000	60.88	16.84	0	80
Age	10,000	49.32	16.23	15	99
Female, proportion	10,000	0.51	–	0	1
Education: ≤ middle school	10,000	0.07	–	0	1
Education: high school	10,000	0.25	–	0	1
Education: graduate school or higher	10,000	0.13	–	0	1
Disability status	10,000	0.03	–	0	1
Monthly income (10K KRW)	10,000	237.55	209.73	0	1,050
Single-person household	10,000	0.17	–	0	1
Older adults (age 65+)	10,000	0.22	–	0	1
Cluster-level Connectivity Index (Connectivity, norm.)	50	0.5	0.18	0.18	0.98
Available for Use (norm.)	50	0.5	0.19	0.12	0.96
Affordability (norm.)	50	0.5	0.22	0.07	1

Note. N = 10,000 individuals nested within j = 25 autonomous districts (2023–2024 pooled). Income is measured in 10,000 KRW units. District-level variables are min-max normalized indices (0–1 scale).

Table 4. HLM Estimation Results

Variable	Model 1	Model 2	Model 3
<i>Level 1: Individual characteristics</i>			
Age (centered)	−0.347*** (0.012)	−0.347*** (0.012)	−0.278*** (0.014)
Female	2.636*** (0.305)	2.634*** (0.305)	2.601*** (0.305)
Education: middle school or below	−22.618*** (0.770)	−22.610*** (0.770)	−19.019*** (2.379)
Education: high school	−7.289*** (0.380)	−7.282*** (0.380)	−7.258*** (0.380)
Education: graduate school or higher	2.906*** (0.483)	2.909*** (0.483)	2.912*** (0.481)
Income (centered)	0.003** (0.001)	0.003** (0.001)	0.002** (0.001)
Job: professional	2.329*** (0.451)	2.326*** (0.451)	2.28*** (0.449)
Job: blue-collar	−0.373 (0.424)	−0.379 (0.424)	−0.385 (0.423)
Job: other	−1.233** (0.456)	−1.226** (0.456)	−1.105* (0.454)
Disability	3.084*** (0.888)	3.081*** (0.888)	3.006*** (0.884)
Single-person household	−1.775*** (0.338)	−1.776*** (0.338)	−0.877* (0.367)
Detached house	−2.478*** (0.377)	−2.480*** (0.377)	−2.574*** (0.379)
Multi-family housing	−1.798*** (0.408)	−1.795*** (0.408)	−1.933*** (0.408)
Year (2024)	0.294 (0.283)	0.292 (0.283)	0.292 (0.283)
Older adults (age 65+)	−6.046*** (1.696)		
<i>Level 2: Autonomous district characteristics</i>			
Connectivity	−1.127 (0.747)	−1.688* (0.804)	
Available for Use (Wi-Fi)	0.085 (0.780)	−0.200 (0.829)	
Affordability	−1.578 (1.135)		
Autonomous district income	−2.035† (1.181)		
Aging rate	−0.298 (1.095)	−0.293 (1.111)	
<i>Interactions</i>			
Older adults × single-person household	−3.958*** (0.936)		

Variable	Model 1	Model 2	Model 3
Age × lower educational attainment	−0.277*** (0.080)		
<i>Cross-level interactions</i>			
Connectivity × older adults	−1.920 (1.901)		
Wi-Fi × older adults	3.026* (1.515)		
Autonomous district income × older adults	2.358 (1.532)		
Connectivity × lower educational attainment	10.936*** (3.370)		

*Note. Unstandardized coefficients with standard errors in parentheses. Model 0 = null model (intercept only). Reference categories: male, college education, white-collar job, apartment, multi-person household. † $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$*

3.3 Platform Text Analysis: LLM-Based Case Analysis of Digital Connectivity

Sections 3.1 and 3.2 quantitatively measured Seoul's digital divide using administrative data and survey data and separated the roles of individual characteristics and regional infrastructure. However, quantitative indicators alone make it difficult to capture when, and in what contexts, residents experience a lack of digital connectivity in everyday life. Qualitative evidence must therefore be integrated to narrow the distance between lived experience in the lifeworld and structural indicators of digital connectivity.

The purpose of this section is to use unstructured text from a local community platform to systematically extract latent cases of digital deprivation across the six dimensions of UMC (Connectivity, Available for Use, Affordability, Devices, Digital Skills, Safety) and to present a methodological framework for combining these cases with the quantitative analyses in Sections 3.1 and 3.2. Rather than treating the platform itself as the object of analysis, the section seeks to recover the connection between structure and experience through the district-level lived-experience data provided by the platform.

3.3.1 Characteristics of the Danggeun Platform

The source of the unstructured text is neighborhood-life posts from Danggeun, a Korean local community platform. Danggeun was selected for three reasons. First, each post is automatically assigned geographic metadata at the administrative-dong and autonomous-district levels, allowing direct linkage with the UMC index constructed in Section 3.1. Second, platform users are distributed broadly across all of Seoul's 25 autonomous districts, reducing concentration in a specific area or age group. Third, because posts voluntarily record digital problems encountered in everyday life, they can reveal lived deprivations that researchers may not capture through surveys. The focus of this study is not to evaluate the platform itself, but to use these structural advantages to collect experiences from the lifeworld.

3.3.2 Why LLMs: Limitations of Existing Methods and the Structural Advantages of LLMs

Classifying unstructured text according to the six dimensions of UMC requires meeting two requirements. First, it is necessary to determine accurately whether a post is related to a digital connectivity problem and, if it is, which dimension it corresponds to. Second, colloquial Korean written in everyday language is difficult to process accurately with simple morphological analyzers because of its agglutinative features, including complex particles and inflectional endings. Existing topic modeling or keyword-based classification is useful for identifying latent themes, but it either does not align with predefined dimensions or has limitations in analyzing conversational text. These limitations are more pronounced because Korean morphological analyzers have low accuracy on colloquial data, while machine-learning models that require data training demand sufficiently large labeled datasets.

Large language models (LLMs) overcome a substantial share of these constraints. Because they are pretrained multilingually, they can grasp context without Korean morphological analysis, and models that handle 104 or more languages simultaneously share cross-linguistic semantic representations that allow them to adapt to new tasks with only a small number of examples. When LLMs are used as tools, their probabilistic outputs and variability must be considered. Because results can differ even for the same input, this study fixed the analytic procedure and used the LLM only as a computational module for classification and inference. This design makes it possible to apply researcher-defined classification rules and inference protocols consistently while drawing on the contextual understanding capacity of the LLM.

3.3.3 Analytic Structure

The text analysis consists of four stages. First, inaccessible posts deleted from the crawled raw CSV data, along with advertising and duplicate posts, are removed, and candidate UMC posts are extracted using keyword dictionaries for the six dimensions. The keyword dictionaries organize terms directly or indirectly related to digital connectivity, and only posts containing those terms are extracted in the first pass. Next, a first-stage classification (`umc_related = Y/N/?`) determines whether each post is related to a digital connectivity problem; to broaden the scope of judgment, the LLM is used without detailed examples or restrictions. Then, only posts confirmed as related are classified into cases corresponding to the six conceptual dimensions by presenting the LLM with a few-shot prompt that includes detailed examples provided by the user. In this stage, the model's autonomy is constrained by providing only concrete classification criteria and specifications, including examples in the prompt, and by strictly managing the context, that is, the information space the model refers to in making judgments.

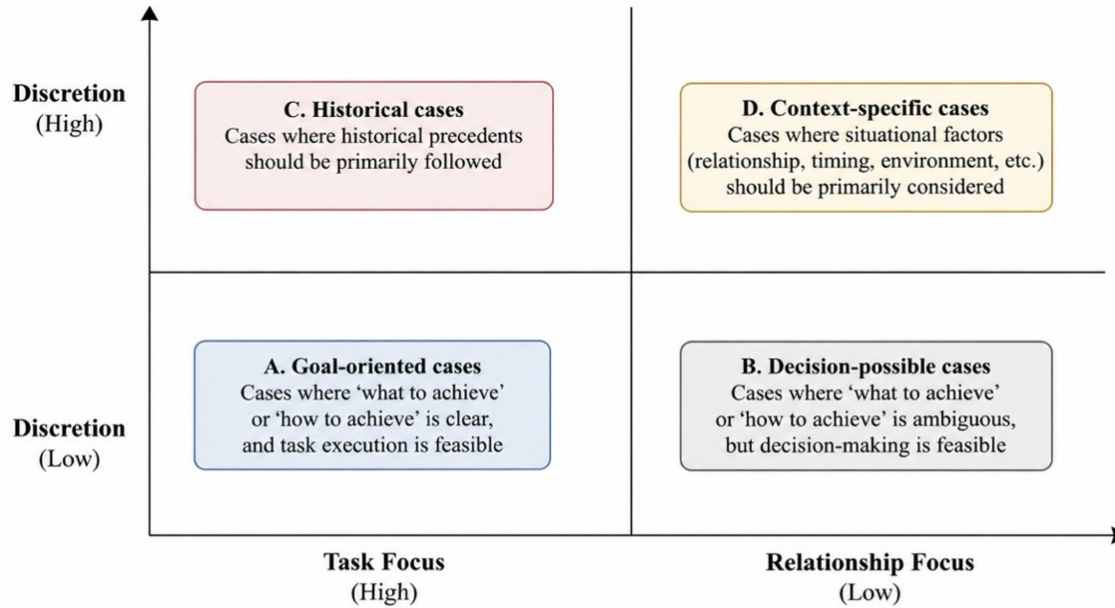
Table 5. summarizes the analytic structure and sampling process. Of the 1,287,761 raw records, approximately 51.5% were excluded from the analysis because their body text could not be viewed due to poster withdrawal or blocking. After duplicate and advertising posts were removed, approximately 480,000 posts remained, and applying the keyword dictionaries for the six dimensions extracted 26,742 UMC-related candidates. In the first-stage judgment using the few-shot prompt, approximately 57.7% (15,438 posts) were classified as related (Y), 29.6% (7,921 posts) as unrelated (N), and 12.6% (3,383 posts) as indeterminate (?). Dimension classification was then conducted only on the posts classified as Y.

In the dimension-classification judgment, the LLM was given prompts containing three to four representative examples for each dimension. Such few-shot prompts help the model learn human judgment criteria and have been shown to perform better than zero-shot prompting on complex tasks (ref). The examples also included negative examples showing which types of posts were unrelated to digital connectivity issues, thereby narrowing relevance as much as possible and securing high precision. In the second-stage dimension classification, only specific classification criteria and definitions were provided in the prompt, so that the model could classify cases more autonomously. In this respect, the earlier stage used an example-based few-shot approach, whereas the later stage used rule-based prompts, showing a balance between reasoning and few-shot prompting (see Figure 6).

Table 5. HLM Estimation Results

Stage	Description	Remaining Number of Posts
Raw Data	All Neighborhood Life posts written between May 2024 and March 2026	1,287,761
Accessible Posts	Removed posts whose body text could not be checked because of account withdrawal or blocking	624,564
Duplicate and Advertisement Removal	Removed duplicate content and advertising or promotional posts	480,972
UMC Keyword Filter	Extracted candidate posts by applying keyword dictionaries for the six dimensions	146,933

Stage	Description	Remaining Number of Posts
First-Stage Judgment	Classified UMC relevance using a few-shot prompt	



Note. The vertical axis represents the degree of discretion in decision making (high vs. low). The horizontal axis represents the focus of decision making (task focus vs. relationship focus). Quadrant A emphasizes goal clarity and task execution, B emphasizes decision-making under ambiguity, C emphasizes adherence to historical precedents, and D emphasizes contextual sensitivity.

Figure 6. Distinguishing the Roles of Classification and Reasoning

Subsequently, the classification results are aggregated at the autonomous district level to create expression profiles, and Bayesian updating is applied to combine them with the quantitative indicators in Section 3.1. Bayesian updating was applied to combine quantitative indicators with text classification results. The prior distribution is the latent space of the six UMC concepts. For autonomous district j in dimension d , let the observed UMC value be s_{jd} . The prior distribution is $Beta(\alpha_{jd}, \beta_{jd})$ with *mean* s_{jd} so that the prior distribution is assumed to satisfy. Using the concentration parameter κ , it is expressed as $\alpha_{jd} = s_{jd} \cdot \kappa$, $\beta_{jd} = (1 - s_{jd}) \cdot \kappa$. In the text classification, when the number of posts assigned to the corresponding dimension is y_{jd} and the total number of classified posts is n_{jd} , the likelihood is expressed as $Binomial(n_{jd}, \theta_{jd})$. The posterior distribution is $\theta_{jd} | y_{jd} \sim Beta(\alpha_{jd} + y_{jd}, \beta_{jd} + n_{jd} - y_{jd})$ is calculated accordingly. The prior concentration κ uses a default value of 20, with sensitivity analyses at 10 and 30. To prevent the posterior distribution in autonomous districts with small numbers of posts from being excessively dominated by the prior, the post count n_{jd} is adjusted by the living population and used as $n_{jd}^* = n_{jd} \times (livingpop_j / mean_livingpop)$.

$$\alpha_{jd} = \kappa \times s_{jd}$$

$$\beta_{jd} = \kappa \times (1 - s_{jd})$$

$$\theta_{jd} | y_{jd} \sim Beta(\alpha_{jd} + y_{jd}, \beta_{jd} + n_{jd} - y_{jd})$$

$$n_{jd}^* = n_{jd} \times (\text{livingpop}_{\text{district}_j} / \text{mean_livingpop})$$

In the Bayesian aggregation, qualitative inference about causal mechanisms was conducted for autonomous district-dimension combinations in which the posterior distribution estimated from posts moved meaningfully away from the prior distribution. Specifically, interpretation was performed only when (1) the posterior shift was statistically significant, (2) there were at least five posts classified as Y in the relevant autonomous district and dimension, and (3) the interpretation remained within the bounds permitted by the multilevel model results in Section 3.2. For the combinations derived from the prior distribution, three to five representative posts were selected, and the linkage pathway between structural indicators and lived experience was described in one or two sentences. Classification and inference were conducted by three independent agents through different reasoning paths, and the final judgment was made by a fourth agent who compared and verified the three results. This followed a triangulation approach that controls bias through cross-validation from multiple perspectives rather than relying on a single protocol.

All classification and inference tasks were conducted by three independent analytical agents using prompts and contexts based on different reasoning approaches, and a fourth analyst compared and verified the three results. This procedure is grounded in the triangulation approach used in qualitative research to reduce bias by drawing on multiple data sources and analysts. A post was included in the final analysis only when the three perspectives converged, and interpretive uncertainty was reflected in the presentation of the results. The results of this process are summarized in <Table 6> and <Figure 7>.

Table 6. . Analysis Units and Interpretation Units in Section 3.3

Category	Unit	What It Shows	Interpretive Cautions
Observation Unit	Individual Post	Clues to experiences of digital connectivity	Regional structure cannot be asserted on the basis of a single post alone
Pattern Unit	Administrative-dong Posts	Expression ratios and distributions by dimension	Aggregation criteria and sample bias must be reviewed together
Structural Unit	UMC Regional Indicators	Regional-level connectivity conditions	Interpret only in a limited manner through consistency with the aggregated text results
Causal-Mechanism Unit	Inference Mechanism	Explains the connection between cases and regional patterns	Interpret only within the bounds permitted by the quantitative results in 3.2

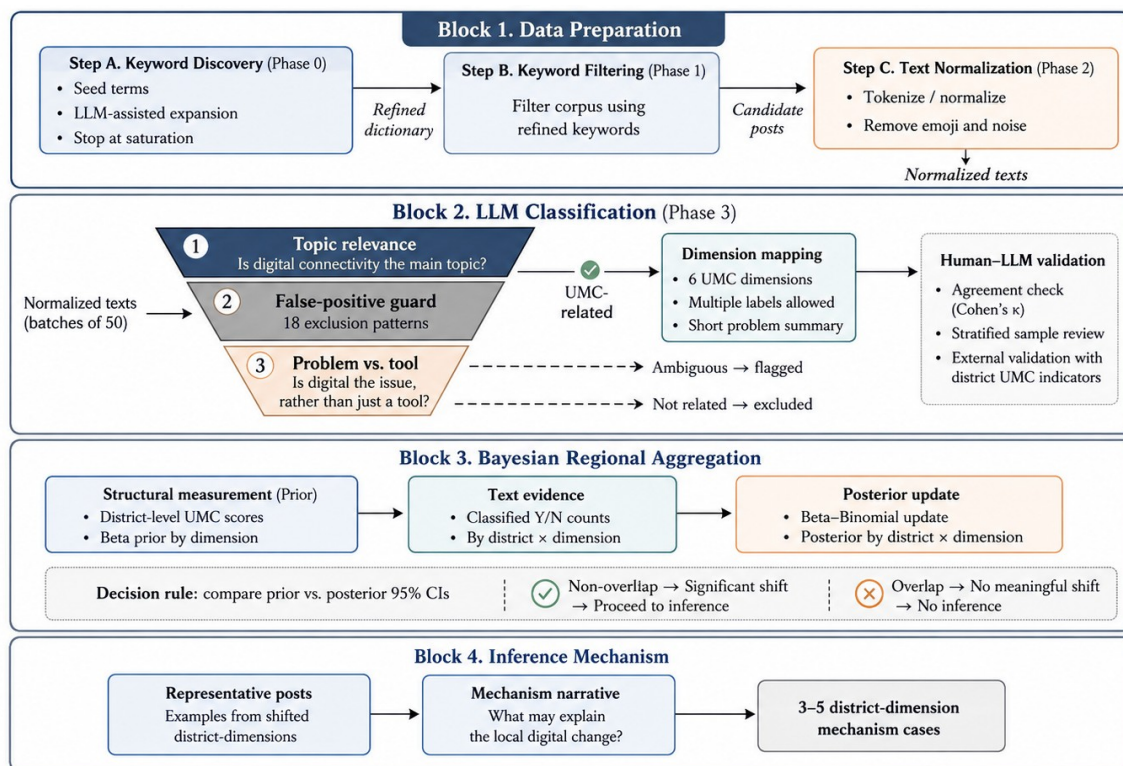
3.3.4 Data and Scope of Application

The raw data consist of a crawl of all Neighborhood Life posts written for Seoul's 25 autonomous districts from May 2024 to March 2026, totaling 1,287,761 posts after duplicate removal. Of these, 766,891 posts (59.6%), comprising 663,163 deleted (DELETED) posts (51.5%) and 103,728 blocked (BLOCKED) posts (8.1%), were excluded from the analysis because their body text could not be checked. The final analytic sample consists of approximately 520,000 posts.

A review of the possibility of systematic bias from sample attrition showed that the standard deviation of removal rates across autonomous districts was 4.6%, indicating that variation across districts was not large. Categories that were central to inferring digital deprivation, such as Life/Convenience (47.0%), Neighborhood Incidents and Accidents (48.4%), and Hospitals/Pharmacies (35.5%), had lower removal rates than categories oriented toward socializing or promotion, such as Neighborhood Friends (86.6%) and Education (69.0%). A substantial share of deletions appears to have resulted from batch deletions associated with user withdrawal. Nevertheless, it cannot be completely ruled out that some deleted posts

included cases related to digital deprivation, or that the act of deletion itself was systematically associated with digital deprivation. This constitutes a limitation of the present analysis.

The structural limitations of the Dangeun data must also be recognized. Because platform users themselves are limited to groups with digital access, the experiences of digitally excluded groups are structurally underrepresented. Since the temporal and spatial distribution of posts is not uniform, sample bias may affect comparisons across autonomous districts. These limitations are constraints of the data source, not constraints of the methodology. In contexts where more direct qualitative data are available, the same pipeline could be applied to produce a more precise analysis.



*Posts are treated as likelihood evidence rather than direct measures of regional UMC.
Inference is made only when posterior shifts are significant and within causal bounds.*

Figure 7. LLM-Based Text Analysis Pipeline

3.3.5 Analysis Results

The text classification results are shown in Table 6. The Connectivity (Connection Quality) dimension has the smallest range of shifts among the six dimensions (-0.006 to +0.005). It can be viewed as a dimension in which measured values and textual evidence generally converge. Yangcheon-gu has a relatively low measured UMC value of 0.278, but its share of related posts is comparatively high, producing a shift of +0.005. By contrast, Yongsan-gu has a high measured value of 0.884, but related posts are rare, producing a shift of -0.006.

In the Availability for Use dimension, Dobong-gu had the lowest measured value (0.213) while also having a high share of related posts, producing a shift of +0.006. Gwangjin-gu had a high measured value of 0.832 and few related posts, recording a shift of -0.010.

The Affordability dimension showed a pattern consistent with income levels across Seoul's autonomous districts. Jungnang-gu and Gangseo-gu each showed positive shifts of +0.005, whereas Seocho-gu had the highest measured value, at 0.993, and few related posts, producing a shift of -0.007. This dimension is consistent with the tendency for concerns about telecommunications fees and device purchase costs to be expressed differentially by income level.

The Devices dimension shows the largest variation across autonomous districts (gap = 0.090). Jungnang-gu had a measured value of 0.052, but its share of device-related posts was 4.95%, the highest among all autonomous districts, producing a shift of +0.048. Songpa-gu also showed a similar pattern, with a shift of +0.026. By contrast, Yongsan-gu (-0.042) and Dongjak-gu (-0.033) had high measured values but few related posts. However, in this dimension, the measured indicator captures device ownership, whereas the textual evidence captures everyday transactions and difficulties related to devices, such as secondhand transactions and repair requests. It is therefore difficult to rule out the possibility that the two indicators partly refer to different objects.

The Digital Skills dimension has the second-largest variation after the Devices dimension (gap = 0.092). Jungnang-gu showed a shift of +0.047, with a measured value of 0.242 and a related-post share of 4.8%. Gangnam-gu also had a relatively low measured value of 0.276 and a comparatively high share of related posts, producing a shift of +0.013. By contrast, Gwanak-gu had the highest measured value, at 0.795, and few related posts, producing a shift of -0.045.

The Digital Skills dimension showed the second-largest variation after the Devices dimension. Jungnang-gu showed a shift of +0.047, with a related-post share of 4.8% relative to a measured value of 0.242, whereas Gangseo-gu showed a shift of -0.031 because related posts were few despite a high measured value. In the Safety dimension, phishing and voice-phishing cases constituted the main content of the posts, and positive shifts were identified in Jungnang-gu (+0.018) and Dobong-gu (+0.015). Jongno-gu had a very high measured value and few related posts, so its posterior distribution shifted downward.

In the Safety and Security (Safety & Security) dimension, Jungnang-gu (+0.018) and Dobong-gu (+0.015) had high shares of posts related to phishing scams and voice phishing. Jongno-gu had the highest measured value, at 0.984, but few related posts, producing a shift of -0.019. It is necessary to also consider the possibility that the residential composition of a tourism and commercial center differs from the characteristics of Dangeun users.

When aggregated at the autonomous district level, differences between regional measured values and post distributions can be classified into three broad types. First are autonomous districts with positive mean shifts. Jungnang-gu showed the largest positive shift in four of the six dimensions and had the highest average value, at +0.020. Dobong-gu and Gangnam-gu also showed positive mean shifts, although their magnitudes were small, at approximately 0.001. In Gangnam-gu in particular, positive shifts were observed in the Availability for Use and Digital Skills dimensions despite the district's high average income, suggesting that heterogeneous groups within the autonomous district may be expressing related issues. Second are autonomous districts with shifts close to zero. Gangbuk-gu (-0.000), Gangdong-gu (-0.001), and Dongdaemun-gu (-0.001) fall into this category. In these districts, positive and negative shifts are mixed across dimensions and tend to cancel out on average. Third are autonomous districts with negative mean shifts. Mapo-gu (-0.014), Yeongdeungpo-gu (-0.013), Gwanak-gu (-0.013), and Seocho-gu (-0.009) fall into

this category. In Gwanak-gu, the population composition centered around university areas and the measured value for Digital Skills may be associated with the rarity of help-seeking posts. Seocho-gu showed particularly large negative shifts in the Affordability and Devices dimensions. In sensitivity analyses that varied the prior concentration κ across 10, 20, and 30, differences in posterior means remained at the level of 0.00001 to 0.00010, indicating no substantive change. This suggests that, after adjustment by the living population, the effective number of observations was sufficient and changes in the weight of the prior did not substantially affect the results.

A point requiring caution in interpretation is that a positive shift does not directly mean that the measured indicator has underestimated the true condition. A positive shift may reflect lived inconveniences that the measured indicator does not capture, but it may also include cultural and demographic factors that make discussion of a particular issue more active in a given autonomous district (ref). In addition, sampling bias among platform users and temporal and spatial imbalances in posts may affect interpretation of the results. Therefore, when linking the direction and magnitude of posterior distributions to policy implications, quantitative and qualitative indicators must be considered together. What is meaningful in the results of this section is that the sign and magnitude of shifts vary systematically by dimension and autonomous district. If only the frequency of Danggeun Market posts is aggregated, platform activity itself may be conflated with estimates of UMC levels. The procedure of entering measured values as the prior distribution and combining text classification results as the likelihood partially separates the portion of observed text frequency attributable to platform activity from the portion attributable to UMC levels. In this sense, the prior-posterior shift functions as an indicator of the extent to which textual evidence adjusts measured values.

However, two constraints must be considered when interpreting the direction of a shift. First, even when a positive shift is observed, it cannot be immediately concluded that the UMC level of the corresponding autonomous district is lower than the measured value. A positive shift suggests the possibility that residents perceive lived conditions not captured by the measured indicator, but the way related topics are expressed on the specific platform of Danggeun Market depends on the composition of users by autonomous district. Second, a negative shift should be read not as meaning that the autonomous district's UMC level is higher than the measured value, but rather as a condition in which there is relatively little incentive for related issues to be expressed on the platform.

A cautious interpretation is warranted. As the contrast between Jungnang-gu and Mapo-gu shows, the interpretation of the shift becomes meaningful when combined with diverse contextual information, including the autonomous district's demographic composition and platform-use characteristics. Second, a negative shift should also be read not as meaning that the autonomous district's UMC level is higher than the measured value, but rather as indicating a condition in which there is relatively little incentive for related issues to be expressed on the platform. In this respect, it is also important to consider why local resources may be lacking in practice yet remain undiscussed on the platform.

Table 7. Post Classification Results

Autonomous District	Total Posts (n)	Y Classification	? Classification	N Classification	Y Rate (%)
Jungnang-gu	5,476	794	28	4,654	14.5
Songpa-gu	9,367	860	99	8,408	9.18

Autonomous District	Total Posts (n)	Y Classification	? Classification	N Classification	Y Rate (%)
Seocho-gu	6,170	434	216	5,520	7.03
Eunpyeong-gu	6,140	429	50	5,661	6.99
Dobong-gu	3,491	233	5	3,253	6.67
Seodaemun-gu	4,500	289	5	4,206	6.42
Gangbuk-gu	3,885	247	31	3,607	6.36
Yongsan-gu	3,012	181	12	2,819	6.01
Yangcheon-gu	4,975	296	7	4,672	5.95
Gangdong-gu	6,254	342	231	5,681	5.47
Yeongdeungpo-gu	5,964	306	5	5,653	5.13
Dongdaemun-gu	4,754	238	29	4,487	5.01
Seongbuk-gu	5,437	266	141	5,030	4.89
Jung-gu	1,351	63	16	1,272	4.66
Gangseo-gu	8,174	371	236	7,567	4.54
Geumcheon-gu	2,298	101	7	2,190	4.4
Jongno-gu	2,267	97	27	2,143	4.28
Dongjak-gu	3,675	150	22	3,503	4.08
Gangnam-gu	10,047	404	459	9,184	4.02
Seongdong-gu	404	16	9	379	3.96
Gwanak-gu	11,665	407	75	11,183	3.49
Mapo-gu	7,974	242	254	7,478	3.03
Guro-gu	4,082	121	96	3,865	2.96
Gwangjin-gu	5,440	147	42	5,251	2.7

Autonomous District	Total Posts (n)	Y Classification	? Classification	N Classification	Y Rate (%)
Nowon-gu	5,040	103	46	4,891	2.04

Figure 8. LLM-based text analysis pipeline

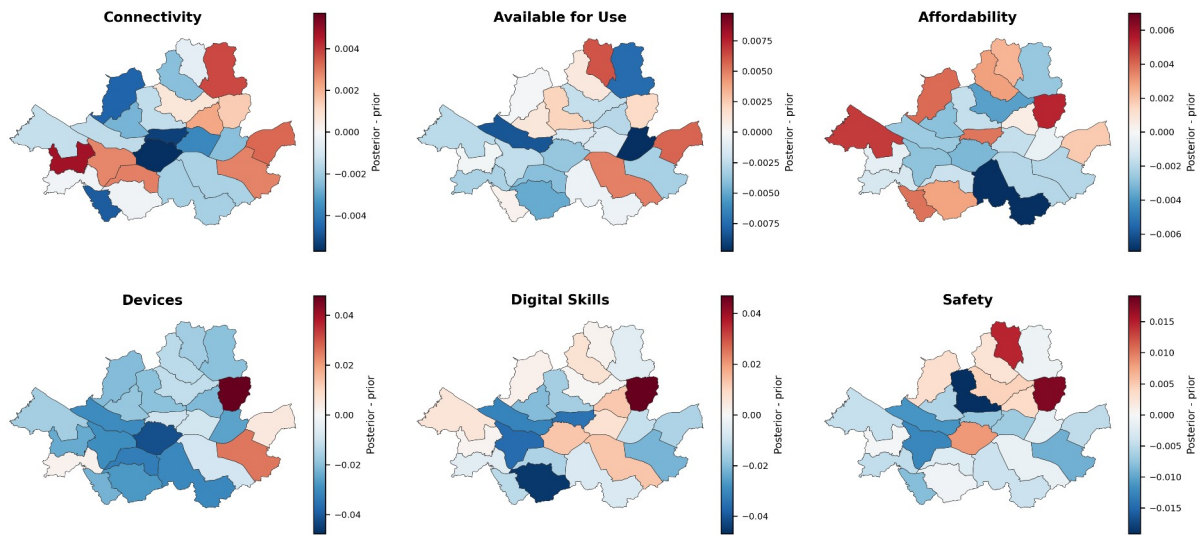


Figure 9. Differences between prior and posterior distributions

Table 8. Hypothesis-generation results: distribution by type of deprivation

Type of deprivation	Related UMC dimension	Number of related posts (n=100)	Share (%)
Information absence	Digital Skills	63	63.0
Lack of institutional pathways	Available for Use	61	61.0
Lack of device access	Devices	50	50.0
Absence of digital competency	Digital Skills	45	45.0
Lack of connectivity infrastructure	Connectivity	40	40.0
Lack of public access points	Available for Use	32	32.0
Lack of economic access	Affordability	27	27.0
Safety and security deficiency	Safety	16	16.0
Composite deprivation (two or more)	—	88	88.0

Note. Because multiple types of deprivation may be inferred from a single post, the sum of the percentages exceeds 100%. Phase 1 is the hypothesis-generation stage, and the final classification is confirmed through triangulation in Phase 2 (judgment).

3.4 Synthesis of Analytical Results

This section synthesizes the results of Section 3.1 (construction of the UMC index), Section 3.2 (HLM multilevel analysis), and Section 3.3 (LLM-based text analysis) to provide an integrated interpretation of the structure through which Seoul's digital divide operates. The three analyses respectively answer the questions of 'where the divide exists' (measurement), 'why the divide arises' (causal identification), and 'how the divide is experienced' (qualitative context), and policy-relevant implications emerge at their points of intersection.

The first finding jointly supported by all three analyses, and the central empirical result of this project, is that Seoul's digital divide is structured by differences between individuals rather than by differences across autonomous districts. In the HLM analysis in Section 3.2, the ICC was only 0.49%, meaning that 99.5% of the variance in digital usage scores was attributable to differences between individuals within the same autonomous district. This result indicates that, although the cross-district variation in the UMC composite index constructed in Section 3.1 is real, its direct effect on individuals' level of digital usage is limited. However, this low ICC does not mean that district-level variables are meaningless; rather, it suggests that their effects are expressed differentially across groups. The cross-level interaction results in Model 3 support this interpretation.

The text analysis in Section 3.3 specifies how this quantitative finding is manifested in everyday life. The fact that 'information absence' appeared as the most frequent type of deprivation (63%) suggests that, among those with lower levels of education, an information gap may intensify in which the very existence of digital services is not recognized. Cases in which users did not know about self-service printing at convenience

stores, did not know the ISP customer center number, or did not know about Government24's public-document printing function all point to a structural problem in which 'services exist but are not visible to users.' This can be interpreted as complementary evidence that explains, at the experiential level, why the education gradient is steep in the HLM.

The second finding is that the effects of autonomous-district infrastructure are not significant at the level of main effects, but operate significantly as moderating effects for specific vulnerable groups. In Model 2, the main effect of the infrastructure variable related to the Connectivity dimension among the UMC dimensions did not reach statistical significance ($p = 0.237$), a result consistent with the limited main effects of district-level variables when the ICC is 0.49%. However, the significant cross-level interaction in Model 3 confirmed that the effect of infrastructure operates not as an average effect, but as a group-specific moderating effect.

The text analysis in Section 3.3 provides the qualitative context for this differential effect. The fact that information absence regarding public access points (print cafes, library multifunction printers, and self-service printing) was inferred in 32% of posts shows that, even when infrastructure physically exists, it is not converted into substantive benefits unless users' awareness and access capabilities are also present. The case from Sinsa-dong, Eunpyeong-gu ('Could it really be that there is none?') suggests the possibility that the user had already attempted an online search but found the results insufficient, thereby highlighting the gap between the existence of infrastructure and its 'discoverability.' When combined with the HLM result that the infrastructure main effect was not significant, this supports the interpretation that the effects of infrastructure investment emerge only when accompanied by support for information access and capabilities among vulnerable groups.

The third finding is that the digital divide is deepened by intersections of complex vulnerability rather than by a single axis. In the HLM, older adults living alone scored an additional 4.0 points lower than older adults not living alone, and the negative effect of lower educational attainment was found to accelerate as age increased. Section 3.3 likewise showed a consistent pattern of composite deprivation. On average, more than 2.5 types of deprivation were identified per post, and when information absence, competency absence, and blocked institutional pathways overlapped, users were found to leave official channels and rely on the community. The absence of a digital support network ('because they do not have children, or because they live apart from them and find it difficult to ask') shows that household composition functions as an informal pathway for acquiring digital competency, and can therefore be interpreted as a mechanism explaining the additional gap among older adults living alone.

The fourth finding concerns the spatial clustering of Digital Deserts and heterogeneity by dimension. In Section 3.1, among the bottom five areas, northern Seoul in particular (Dobong-gu, Gangbuk-gu, Nowon-gu, and Jungnang-gu) formed a Low-Low cluster in the UMC composite index; while these areas recorded above the city average on the Connectivity dimension, they showed extreme deficits in the Devices and Safety dimensions. This heterogeneity in dimension-specific profiles suggests the need for dimension-specific tailored interventions rather than uniform infrastructure investment.

From a methodological perspective, combining the three analyses allows their respective limitations to be mutually compensated. The composite index in Section 3.1 identifies the relative positions of autonomous districts but cannot capture within-individual variation, while the HLM in Section 3.2 separates the contributions of individuals and areas but cannot explain the qualitative pathways through which the digital divide is manifested. The text analysis in Section 3.3 visualizes the experiential context of deprivation but is limited in representativeness. By combining the three analyses through sequential triangulation rather than in parallel (measurement → causal identification → qualitative context), the project was able to construct an integrated answer to the three questions of 'where' (autonomous districts), 'why' (the interaction between individual characteristics and infrastructure), and 'how' (the pathways through which deprivation is manifested).

These synthesized results lead directly to the policy recommendations in Chapter 4. The finding that individual characteristics are the dominant explanatory variables provides the basis for prioritizing people-based policies; the differential effects of infrastructure support the need for place-based policies targeted to vulnerable groups; the confirmation of complex vulnerability indicates the need for an integrated support

system rather than a single program; and the identification of spatial clustering provides the basis for subregional governance.

4. Policy Recommendations

This chapter presents policy directions for addressing Seoul's digital divide based on the analytical results in Chapter 3. The data used in the analysis were collected in 2023-2024 and therefore reflect the policy environment under the Framework Act on Intelligent Informatization. As of January 2026, the Digital Inclusion Act came into effect, shifting the policy frame from the previous approach centered on closing the information gap to the expanded concept of digital inclusion. The recommendations in this chapter present the gaps and vulnerabilities identified under the previous system as empirical evidence, while exploring their potential policy application within the new legal framework.

4.1 Theoretical Framework for Policy Recommendations

Policy approaches to Seoul's digital divide can be divided into two axes, people-based and place-based, and the analytical results in Chapter 3 suggest that the former is appropriate as the central axis of policy. In Section 3.2, Model 1, which included only individual-level variables, explained 83.4% of the variance between autonomous districts; this means that most of the cross-district gap in digital usage is attributable less to intrinsic local characteristics than to differences in the sociodemographic composition of residents in each district. Accordingly, the primary route of policy intervention should be located in a people-based approach that directly strengthens the digital competency of groups with multiple vulnerability factors, such as older adults with lower educational attainment and older adults living alone.

However, the place-based approach here must be distinguished from uniform resource allocation at the autonomous-district level. The ICC of 0.49% indicates that variance between autonomous districts is not the main source of the digital usage gap, suggesting that policies centered on area-level allocation may be inefficient. What this project proposes is a place-sensitive approach: a strategy that amplifies the effects of people-based policies by improving spatial conditions. As confirmed in the cross-level interactions, the fact that public Wi-Fi density had a disproportionate effect for older adults (+3.03 points), and that the level of connectivity had a disproportionate effect for those with lower educational attainment (+10.94 points), means that raising the infrastructure level in areas with high concentrations of vulnerable groups can function as a key multiplier for people-based policies.

This dual approach is also consistent with internationally accepted policy frameworks. OECD(2023) emphasizes that, under a structure of widening interregional gaps (uneven divergence), place-based policies and people-centered transfer policies are not mutually exclusive but complementary, while ITU(2024)'s UMC policy agenda likewise recommends both target-group-centered policies and the development of national digital strategies and roadmaps. EU Digital Decade 2030 establishes national roadmaps for each member state across four axes - digital skills, infrastructure, business, and public services - and operates a cyclical structure in which progress is reviewed through annual monitoring (European Commission, 2025). This structure of measurement -> monitoring -> policy adjustment partly aligns with the analytical procedure of this project.

4.2 People-based Policy Recommendations

(1) Older adults living alone

The predicted digital usage gap for older adults living alone is approximately -10.87 points, corresponding to 13.6% of the full scale. Whereas older adults in multi-person households can receive informal digital support from household members, this pathway is absent for older adults living alone. The text analysis in Section 3.3 confirmed how this gap is manifested in everyday life. The case of an IT planner in Munjeong-dong, Songpa-gu, who proposed app-use training for middle-aged and older adults by stating, 'I suppose those reading this post are already using them reasonably well; for those who do not have children, or who live apart from them and find it difficult to ask,' visualizes the structure through which the absence of an in-household digital support network leads to persistent difficulties with everyday digital tasks.

Existing Digital Learning Centers have centered on group education and have faced structural limitations in physically reaching older adults living alone whose mobility is constrained. With the transition to AI Digital Learning Centers from March 2026, outreach dispatch education is expected to expand (Ministry of Science and ICT, 2025.12.24), but program designs specialized for older adults living alone have not yet been specified. The United Kingdom's Age UK Digital Champion Programme has reported improvements in older adults' digital confidence by combining volunteer-based one-on-one home visits with device lending (Age UK, 2023).

Taken together, policy for older adults living alone is insufficient if it only expands access to group education; what is needed is a model that substitutes for social scaffolding. Specifically, this report recommends a one-on-one home-visit support system through the cultivation of dong-level Digital Companions, the operation of cohorts tailored to older adults living alone, and the parallel provision of device lending.

(2) Older adults with lower educational attainment

In M3, the age x lower educational attainment interaction (-0.28 , $p < 0.001$) shows that the age-specific slope of decline in digital usage for the lower-educational-attainment group is approximately twice as steep as that for the college-graduate reference group. Over a 20-year age range, this difference accumulates into an additional gap of approximately 5.5 points, suggesting that the digital divide between educational groups tends to widen as age increases.

The text analysis in Section 3.3 shows how this education gradient is manifested in everyday life. The fact that 'information absence' appeared as the most frequent type of deprivation (63%) suggests that, as educational level decreases, the gap in awareness of the very existence of digital services intensifies. Cases in which users did not know about self-service printing at convenience stores, did not know the ISP customer-service telephone number, or did not know about Government24's public-document printing function all point to a condition in which 'services exist but are not visible to users.' Such information absence may be strongly associated with educational level, because informational skill is proportional to level of education (Hargittai, 2010).

It is unclear whether existing Digital Learning Centers systematically provided differentiated pathways by educational level. Singapore IMDA's Seniors Go Digital operates an integrated model that establishes a three-stage pathway of basic, intermediate, and advanced levels, embeds cybersecurity education at each stage, and combines device and fee subsidies for low-income older adults (IMDA, 2020; CSA Singapore, 2023). Security education is particularly relevant in the Seoul context, where vulnerability in the Safety dimension was identified in Section 3.1.

The recommendations are to systematically differentiate staged pathways by educational level, to adopt an integrated design that makes security education mandatory at each stage, and to link device and fee support for low-income older adults with lower educational attainment.

(3) Strengthening the discoverability of digital services to address information absence

One policy task raised by the text analysis in Section 3.3 is that existing digital competency education has focused on 'how to use' services, whereas the gap in 'awareness of existence' has not been sufficiently addressed as a policy issue. Information absence, inferred in 63 of the 100 cases analyzed, refers not to a lack of capability to use services but to a condition in which the existence of the services themselves is unknown.

For example, in the case of a Bongcheon-dong resident in Gwanak-gu who asked, 'It seems that court documents and the like cannot be printed at ordinary print cafes; does anyone know where official documents can be printed?', neither the Government24 official-document printing function nor the existence of court self-service issuance kiosks was visible to the user. The expression 'Is there truly none?' used in Sinsa-dong, Eunpyeong-gu suggests that the user may already have attempted an online search but found the results insufficient, highlighting the gap between the existence of infrastructure and its 'discoverability.'

This absence of information is difficult to resolve simply by strengthening publicity. Passive provision of information reaches only users who already possess the capacity to search for information. Therefore, as a policy design principle, an approach is needed that embeds the discoverability of digital services from the

service design stage onward. Specifically, we propose functions in public service apps (such as Seoul ON) that provide location-based guidance to nearby public digital access points (self-service printing, library multifunction printers, and AI Digital Learning Centers), offline linkage that informs residents of relevant digital services when they visit community service centers for civil affairs, and the parallel listing of relevant digital channels in text-message notifications from the competent dong office.

(4) Structural Responses to the Educational Gradient

Educational attainment is the strongest predictor of digital use, and this gap has a structural character that is difficult to resolve through short-term programs alone. It is therefore necessary to distinguish the time horizons of policy. In the short term, the targeted programs described above can contribute to mitigating current gaps; in the medium to long term, systematically incorporating digital competency into curricula from the basic education stage becomes a structural pathway for preventing the cumulative widening of gaps in future generations. The 'matters concerning activities to cultivate digital competency' and 'education, counseling, and publicity for digital inclusion' required in the basic plan under Article 8 of the Digital Inclusion Act provide the institutional basis for encompassing precisely these short-, medium-, and long-term policies at the same time.

4.3 Place-Based Policy Recommendations

(1) Dimension-Specific Tailored Interventions for Digital Desert Autonomous Districts

Section 3.1 classified the five autonomous districts at the bottom of the composite index (Jungnang-gu, Dobong-gu, Gangbuk-gu, Guro-gu, and Nowon-gu) as 'Digital Deserts.' Their core characteristic is that, while they record levels at or above the city average in the Connectivity dimension (Jungnang-gu 0.630, Gangbuk-gu 0.562), they exhibit extreme deprivation in the Devices (Jungnang-gu 0.052) and Safety (Dobong-gu 0.019) dimensions. This suggests that the main source of disparity is not the absence of physical network infrastructure, but rather a lack of the human and economic capabilities needed to use it.

Because the dimension-specific decomposition (Figure 5) confirmed that each autonomous district has different vulnerable dimensions, dimension-specific tailored interventions may be more effective than uniform support. For example, device distribution and lending programs should be prioritized in Jungnang-gu, where the Devices dimension is extremely low, while security education and awareness-raising programs should be prioritized in Dobong-gu, where the Safety dimension is effectively close to zero. For autonomous districts vulnerable in the Affordability dimension, fee subsidies and expansion of low-cost plans constitute a more direct intervention pathway.

The text analysis in Section 3.3 provides concrete directions for dimension-specific tailored interventions. In the case of Dobong-gu, where the Safety dimension is extremely low, the text analysis showed that cases of harm in online secondhand transactions were a major manifestation of safety deprivation. The pattern in which the absence of capacity to verify the condition of secondhand electronic devices is combined with a lack of information about formal redress channels after harm occurs (consumer counseling centers, police reports, etc.) suggests that security education should extend beyond simple password management to include consumer protection capacities in digital transaction environments.

Given that the main effect of autonomous-district infrastructure variables in M2 was not significant overall ($p = 0.237$), place-based interventions have limits as a standalone means of closing the gap. This means that more substantive effects can be expected when dimension-specific tailored interventions are combined with the people-based policies discussed in 4.1.1, for example, when device distribution is accompanied by level-specific education.

It is necessary to examine whether the current policy system allocates resources differentially in ways that reflect the characteristics of each autonomous district. The United Kingdom's Digital Inclusion Innovation Fund (£11.7M, 2025) adopted a model that allocates funding to 80 local projects through a community-led approach while differentiating project content according to local needs (older adults, low-income groups, migrants, etc.) (UK DSIT, 2025). This provides a reference case demonstrating the feasibility of differentiated interventions suited to local contexts.

(2) Redesigning Public Digital Access Points

The inference, in the text analysis in Section 3.3, that 32% of posts reflected a lack of public access points raises the possibility that Seoul's public digital infrastructure is not quantitatively insufficient, but rather is designed and located in ways that do not correspond to users' task contexts. In cases such as a student living alone in Sillim-dong, Gwanak-gu asking for a recommendation for 'a PC room where document scanning is possible because mine is not working,' or a Bongcheon-dong resident in Gwanak-gu asking, 'Is there anywhere near Sadang Station where I can use the internet and a printer?', users were not aware of existing public access points such as self-service printing in convenience stores or library multifunction printers.

These findings suggest that policy on public digital access points should shift from 'installation' to 'discoverability and accessibility.' Specifically, we propose establishing an integrated guidance system for public digital access points by autonomous district (self-service printing, scanning, public Wi-Fi, and AI Digital Learning Centers), strengthening searchability through data linkages with private platforms such as Naver Map and KakaoMap, and securing access points outside the operating hours of public facilities, including late-night and weekend periods (for example, 24-hour self-service printing booths).

(3) Coalition Governance for Seoul's Northern Region

The spatial distribution of the UMC composite index (Figure 1) and the LISA cluster analysis (Figure 5) show that Digital Desert autonomous districts are clustered in northern Seoul in a statistically significant manner. The physical adjacency of Dobong-gu, Gangbuk-gu, Nowon-gu, and Jungnang-gu, which belong to the Low-Low cluster, suggests that this is not a problem of individual autonomous districts but a structural pattern at the regional scale.

This spatial clustering provides three grounds for recommending cooperative governance at the regional scale. First, the LISA analysis confirmed significant spatial autocorrelation, statistically supporting the interpretation that this is a structural phenomenon rather than a merely visual pattern. Second, if individual autonomous districts independently build digital education programs or infrastructure, resources become dispersed; however, if neighboring autonomous districts jointly operate digital hubs or share education programs, economies of scale can be achieved. Third, as confirmed in Section 3.1, adjacent autonomous districts differ in their dimension-specific strengths and weaknesses (e.g., Jungnang-gu's strength in Connectivity at 0.630 versus its weakness in Devices at 0.052), making cross-utilization of resources possible.

Specifically, we propose establishing coalition governance for the northern Digital Desert region (Dobong, Gangbuk, Nowon, Jungnang, and others). This includes operating joint digital education hubs, cross-utilizing dimension-specific strengths and weaknesses, and securing budgets and planning projects at the regional scale. The institutional pathway for this regional cooperation system is discussed in 4.1.3 in connection with the basic-plan system of the Digital Inclusion Act.

4.4 Integrated Policy Recommendations

The preceding analytical results and policy recommendations suggest that the most substantive effects can be expected when people-based and place-based approaches are combined. The combination of spatial targeting and group targeting, that is, prioritizing people-based policies such as outreach support for older adults living alone and level-specific education programs in Digital Desert autonomous districts, constitutes a concrete integrated pathway. The text analysis in Section 3.3 demonstrates the specific form of this integration. The finding that compound deprivation (88%) is widespread suggests that single-axis policy interventions, such as providing devices only, offering education only, or expanding infrastructure only, are insufficient to address the overlapping structure of deprivation. For example, even if devices are provided, the effects of device distribution will be limited if users do not know that accessible services exist (information absence) or if they lack the capacity to operate the digital interfaces of those services (capacity absence).

Article 8 of the Digital Inclusion Act stipulates that the government must establish a basic plan for digital inclusion every three years, and that this basic plan must include 'improving accessibility and supporting

social participation for digitally vulnerable groups,' 'analyzing the status of digital inclusion and policy performance,' and 'activities to cultivate digital competency,' among other matters. Article 9 of the same Act requires relevant central administrative agencies and local governments to establish and implement annual implementation plans. The results of the multilevel analysis conducted in this project, including the decomposition of between-district and within-district variance, identification of vulnerable groups, and dimension-specific profiling, can be used as empirical evidence in the process of formulating this basic plan. In particular, the distribution by deprivation type derived from the text analysis in Section 3.3 (information absence 63%, absence of institutional pathways 61%, capacity absence 45%) provides empirical evidence that can inform priority-setting in the basic plan.

The institutionalization of the coalition governance for the northern Digital Desert region proposed in 4.1.2 can also be explored within this system. Although the Digital Inclusion Act designates metropolitan and basic local governments as the actors responsible for establishing and implementing implementation plans, it does not contain separate provisions for regional cooperation systems among adjacent basic local governments. In the process of formulating the basic plan, it is necessary to review institutional pathways for such regional cooperation, for example, the creation of consultative bodies for joint project planning among autonomous districts.

Finally, from the perspective of procedural generalizability discussed in 1.1, the analytical framework of this project, namely digital divide diagnosis (construction of a composite index and dimension-specific decomposition) → multilevel analysis (separation of individual and district contributions and identification of vulnerable groups) → extraction of qualitative context (LLM-based classification of deprivation types) → targeted policy design (integration of people-based + place-based approaches), is significant as a policy-derivation procedure that can be applied not only to Seoul-specific prescriptions but also to other cities with similar data structures.

5. Conclusion

5.1 Summary

This project applied the ITU's Universal Meaningful Connectivity (UMC) framework to Seoul's 25 autonomous districts and constructed an integrated analytical framework for measuring, analyzing, and interpreting the digital divide within the city. By combining three stages of analysis, construction of the UMC composite index (3.1), HLM multilevel analysis (3.2), and LLM-based text analysis (3.3), as a sequential triangulation, the project clarified at multiple levels where Seoul's digital divide exists, why it emerges, and how it is experienced.

First, the UMC composite index analysis found a gap of up to 2.5 times in digital connectivity across Seoul's autonomous districts. The difference between Seocho-gu (0.695) and Jungnang-gu (0.279) shows that, contrary to the general perception of Seoul as a highly connected city, a considerable relative gap is observed at lower-level administrative units. In particular, Digital Deserts are spatially clustered in northern Seoul (Dobong-gu, Gangbuk-gu, Nowon-gu, and Jungnang-gu), and these districts exhibit dimension-specific heterogeneity: although they are at or above the city average in the Connectivity dimension, they show extreme deprivation in the Devices and Safety dimensions.

Second, the HLM analysis confirmed that Seoul's digital divide is structured overwhelmingly by individual characteristics rather than by regional infrastructure differences. The ICC was only 0.49%, indicating that 99.5% of the variance was attributable to differences among individuals, and educational attainment (middle school graduation or less, -22.6 points) and age (-0.35 points per year) were the strongest predictors. However, cross-level interactions showed significant effects for public Wi-Fi density × older age (+3.03 points) and Connectivity × lower educational attainment (+10.94 points), confirming that regional infrastructure provides disproportionately large benefits to vulnerable groups. The additional gap for older adults living alone (-4.0 points) and the accelerated pattern of age × lower educational attainment support the existence of compound vulnerability.

Third, the LLM-based text analysis ran three independent inference agents in parallel on 100 Danggeun neighborhood-life posts and generated 388 hypotheses of digital deprivation. Information absence (63%), absence of institutional pathways (61%), and lack of device access (50%) emerged as the main types of deprivation, and two or more forms of deprivation overlapped in 88% of posts. This made visible, at the case level, the manifestation pathways of deprivation that quantitative analysis cannot capture: at what stage and for what reason official channels are abandoned.

Based on these analytical results, Chapter 4 presented policy directions that integrate people-based approaches (older adults living alone, older adults with lower educational attainment, and responses to the educational gradient) and place-based approaches (dimension-specific tailored interventions for Digital Deserts and coalition governance for the northern region). These directions are linked to the basic-plan formulation system under the Digital Inclusion Act, which came into effect in 2026, and the analytical framework itself has procedural generalizability applicable to other cities with similar data structures.

5.2 Limitations and Future Tasks

This project has the following limitations, which indicate directions for future research.

First, there is the limitation of cross-sectional data. The Seoul Survey (2023–2024) used in the HLM analysis and the administrative data used to construct the UMC index capture cross-sections at specific points in time and therefore cannot track temporal changes in the digital divide. Panel data need to be constructed in order to determine how the effects of individual characteristics such as educational attainment or age on digital use change over time, and whether gaps actually narrow after policy interventions. Although the fact that the Seoul Survey is an annually conducted administrative survey suggests the possibility of conversion to a

panel structure, the current sample design is a repeated cross-sectional structure and therefore has limits for individual-level tracking.

Second, there are constraints on causal identification. HLM estimates the direction and magnitude of relationships among variables, but because the analysis is based on observational data, caution is required for strict causal interpretation. For example, it is difficult to separate whether the positive effect of public Wi-Fi $\times$ older age observed in the cross-level interaction is a causal effect of public Wi-Fi or an effect mediated by other characteristics of autonomous districts with high public Wi-Fi density (welfare infrastructure, population composition, etc.). Future research needs to identify causal relationships more rigorously through quasi-experimental designs (such as difference-in-differences) that use policy changes as exogenous shocks.

Third, there are limits to the representativeness of the text analysis. Posts on Danggeun's Neighborhood Life board are limited to platform users, that is, groups that already have digital access; as a result, the experiences of those most severely affected by digital exclusion are structurally underrepresented. In addition, the current analysis is based on the results of Phase 1 (hypothesis generation), which examined 100 posts, while Phase 2 (triangulation and final classification by adjudication agents) has not yet been conducted. Once the future full-corpus analysis (approximately 50,000 posts) and Phase 2 adjudication are completed, it should be possible to derive a more systematic distribution of deficits by autonomous district and by dimension.

Fourth, there are methodological constraints in constructing the UMC index. This project applied equal weights to the six dimensions, but the relative importance of dimensions may vary by context. For example, in Seoul, where device ownership has become widespread, further review is needed as to whether assigning the same weight to the Devices dimension as in other cities is appropriate. In addition, because Min-Max normalization is sensitive to extreme values, additional verification of index stability is required given the small number of autonomous districts (25).

Fifth, there is a limitation in the spatial resolution of autonomous district-level analysis. Autonomous districts ($n=25$) are a relatively coarse unit for capturing spatial variation in the digital divide within Seoul. Even within the same autonomous district, differences may exist across administrative dong, and the administrative-dong metadata assigned to posts in the text analysis in Section 3.3 suggests the possibility of more fine-grained spatial analysis. If administrative-dong-level data become available in the future, analysis with higher spatial resolution will be possible.

Finally, the analytical framework of this project—constructing a composite index based on the UMC framework, separating individual- and district-level contributions through a multilevel model, and extracting qualitative context using LLMs—has value not only as a basis for Seoul-specific prescriptions, but also as a methodological framework applicable to other cities with similar administrative data structures and digital platform ecosystems. To test the external validity of this framework, a future task is to apply it to regions with different levels of urbanization and digital infrastructure environments.

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